

AGROLATTICE

Global Agricultural Intelligence Platform

Detailed User Manual

Release 10.3

Field-by-field instructions, worked examples, output interpretation, agricultural terminology and troubleshooting

Application version	18.4-release10.3-agrolattice-entrypoint
Main entry point	agrolattice.py
Country storage	Datasets\countries\<country>\agroclimate_longformat.csv
Manual date	July 2026

Designed for agronomists, agricultural researchers, field technicians and data scientists

Contents and navigation map

This manual follows the consolidated Release 10.3 navigation. Every original analytical tool remains accessible through its workspace and through All Tools.

Chapter	Workspace	Functions explained
1	Home	Detailed command centre
2	Fields & Operations	Farm portfolio & mapped fields; Tasks, scouting & operations; Sensors, irrigation & nutrition; Crop health, rules & alerts; Management zones & prescriptions
3	AgroLattice Twin	AgroLattice live twin; AgroLattice scenario studio; AgroLattice adaptive copilot; AgroLattice twin configuration
4	Climate & Earth Observation	Climate similarity; Compare two cities; Climate analogues; Explain similarity; Weight sensitivity; Agroclimatic clusters; Spatial intelligence; Composite agricultural risk; Trend analysis; Climate anomalies; Drought & water stress; Climate hazards laboratory; Satellite crop monitoring
5	Crop Decisions	Daily weather & phenology; Seasonal suitability; Suitability stability; Crop planning laboratory; Current-season monitor; Soil-water balance; Yield calculator; Water productivity & economics; AquaCrop simulations; DSSAT & APSIM interoperability
6	Experiments	Maize flowering trials & field data; Maize synchrony analysis & prediction; Projects
7	Models & Evidence	Validation Centre; Model ensemble & disagreement; Yield validation; Statistics toolbox; Scientific robustness
8	Reports	Study & publication builder; Workflow & reporting
9	Data & Settings	Country workspace; Dataset updater; Validated crop parameters; Data exchange, backup & audit; ChatGPT settings; System diagnostics
10	Help	Quick guide & glossary; Commercial feature benchmark

How to use the control reference

Each tool chapter lists tabs, fields, selectors, uploads, action buttons and downloads extracted from the Release 10.3 source. Controls that appear only after another choice are described even if they are not visible at first.

1. About AgroLattice

AgroLattice is a desktop Streamlit platform that connects country climate datasets, NASA POWER weather, Sentinel-2 imagery, crop and soil-water models, mapped fields, field trials, machine learning, validation and reproducible reporting.

Decision-support scope

AgroLattice supports research and transparent screening. It does not replace local agronomic expertise, calibrated instruments, laboratory testing, regulatory approval or independent validation.

1.1 Intended users

- Agronomists and crop scientists who need climate, phenology, water and field-monitoring tools.
- Agricultural researchers and doctoral students designing trials and validating predictive models.
- Field technicians collecting observations, sensor readings and operational records.
- Data scientists who need reproducible agronomic features, grouped validation and model comparison.
- Administrators maintaining country datasets, backups and optional dependencies.

1.2 Starting and verifying the application

1. Double-click RUN_APP.bat. The launcher starts Streamlit using agrolattice.py.
2. If the app does not start, run DIAGNOSE_APP.bat and review the missing dependency or path message.
3. Run VERIFY_RELEASE10_3_AGROLATTICE_NAMING.bat after an update to confirm the correct entry point and version.
4. Install general dependencies with INSTALL_DEPENDENCIES.bat.
5. Install satellite dependencies with INSTALL_SATELLITE_DEPENDENCIES.bat when Sentinel-2 functions are needed.
6. Install AquaCrop-OSPy with INSTALL_AQUACROP_OSPY.bat when process-based AquaCrop simulations are needed.
7. Use CHECK_EXTERNAL_MODELS.bat to inspect DSSAT and APSIM availability.

1.3 Standard data storage

Every country uses the same folder pattern. Mexico is no longer a special top-level dataset.

Datasets\countries\<country>\agroclimate_longformat.csv

Example: Datasets\countries|mexico\agroclimate_longformat.csv

- The country selector determines which country dataset is active.
- Country updater jobs, backups and NASA caches are separated by country.
- worldcities.csv supplies the country, city, administrative-region and coordinate catalogue.
- Do not manually mix country rows into one climate file.

2. Understanding the interface

2.1 Global workspace controls

Control	Meaning	Example
Country	Selects the active country dataset and location catalogue.	Mexico
Prefer results inside selected country	Restricts map-search results to the selected country when possible. Untick	Enabled

Control	Meaning	Example
	for worldwide search.	
Show place search on maps	Shows the map search control on Folium maps.	Enabled
System status	Opens a compact readiness panel for historical climate, daily weather, satellite data and the last page error.	System - Ready
Workspace preset	Changes suggested shortcuts for Agronomist, Researcher, Field technician, Administrator or All tools. It does not remove functions.	Researcher
Find a tool	Searches all 51 original tools by name or concept.	Search “clustering”

2.2 Active workspace context

The context bar keeps the same country, project, field, season, crop and trial available across workspaces. Use Change context to avoid reselecting the same entities repeatedly.

Important

A page can still contain its own location or crop selector. The page-level choice controls that analysis; the context bar controls the wider project and field context.

2.3 Workspace view, Analysis selector and tabs

- Workspace view: chooses a major view inside one consolidated sidebar destination.
- Analysis: chooses a related scientific tool inside a studio, such as Climate Comparison Studio.
- Tabs: divide one tool into stages or result views. Switching tabs does not delete results already stored in session state.
- Expanders: hide advanced options, methods, raw data or diagnostics until needed.
- Primary buttons: run the main calculation. Always review warnings before running.
- Download buttons: export results, templates or complete packages. Save them with the project and analysis date.

2.4 Maps

- Roads & places: OpenStreetMap-style basemap for roads, settlements and labels.
- Satellite imagery: visual basemap for identifying field boundaries and landscape features.
- Light map: low-contrast basemap for overlays and figures.
- Search: enter a place, address, research station or landmark.
- Measure: measure length in metres/kilometres and area in square metres/hectares.
- Drawing tools: draw polygons or rectangles for fields and plots; live edge, perimeter, width, height and area measurements support accurate layout.
- Layer control: turns basemaps and overlays on or off.

2.5 Common status messages

Status	Meaning
Ready	Required data or dependency is available.
Run required	The module is installed but no current-session result exists.

Status	Meaning
Optional	The feature is not required for the main workflow.
Unvalidated	No independent validation result is currently loaded.
Warning	The analysis can continue, but assumptions or data limitations need attention.
Error	The calculation stopped. Read the friendly error and System diagnostics before retrying.

3. Recommended end-to-end workflows

Create a country dataset

1. Select the country.
2. Open Data & Settings > Dataset updater.
3. Choose locations, dates and variables.
4. Download batches.
5. Finalise and audit the candidate.
6. Install after backup and checksum verification.

Map and monitor a field

1. Create a farm and field boundary.
2. Search the map and draw or upload the field.
3. Open Satellite monitoring.
4. Use the mapped field as the AOI.
5. Process Sentinel-2 scenes.
6. Review indices, anomalies and model integration.

Run a maize flowering trial

1. Create the trial and parent lines.
2. Define sowing offsets, blocks and replicates.
3. Draw the experiment field and plots.
4. Randomise treatments.
5. Record daily flowering, height and stress.
6. Record flowering dates, ear traits and seed quality.
7. Run synchrony, prediction and offset optimisation.

Build a field twin

1. Link a mapped field and optional maize trial.
2. Load weather, root-zone, satellite and sensor data.
3. Review the live state.
4. Save snapshots.
5. Compare scenarios.
6. Use the copilot to prioritise measurements.

Validate and publish

1. Assemble observed and predicted data.

2. Use grouped validation.
3. Inspect error, calibration and residuals.
4. Compare models or ensembles.
5. Run robustness checks.
6. Build a publication package with traceable outputs.

4. Home

The Home workspace is the starting point. It summarises readiness, open tasks and alerts, and offers shortcuts based on the selected workspace preset.

Workspace selector

Open Home from the sidebar. Use the Workspace view selector and, where present, the Analysis selector to move between related functions without leaving the workspace.

4.1 Detailed command centre

Path: Home > Detailed command centre

Shows the current workspace context, readiness of major data sources, open tasks, open alerts and recommended next actions.

Worked example

Example: after opening a 2027 maize trial, Home may show three overdue field measurements, one satellite alert and a shortcut to the Adaptive Copilot.

Typical workflow

1. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

This view is primarily informational or inherits its controls from the global sidebar and active workspace context.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

5. Fields & Operations

This workspace is the operational system of record for farms, fields, tasks, scouting, sensors, agronomic operations, alerts and precision-management outputs.

Workspace selector

Open Fields & Operations from the sidebar. Use the Workspace view selector and, where present, the Analysis selector to move between related functions without leaving the workspace.

5.1 Farm portfolio & mapped fields

Path: Fields & Operations > Farm portfolio & mapped fields

Organise farms, mapped field boundaries, crop rotations, local workflow roles and complete field histories.

Worked example

Example: create a research farm, search for its location, draw a 5 ha field, switch between road and satellite basemaps and store the boundary.

Tabs and views

Tab/view	Purpose
Farms	Opens the farms stage or result view.
Fields	Opens the fields stage or result view.
Crop history	Opens the crop history stage or result view.
Team & access	Opens the team & access stage or result view.
Portfolio map	Opens the portfolio map stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Field, Notes, Boundary source, GeoJSON containing one Polygon or MultiPolygon.
3. Run “Assign field access”.
4. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Field	Drop-down	Options: generated from available data	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Notes	Long text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Boundary source	Single choice	Options: Draw on map, Upload GeoJSON, Current project, Satellite crop monitoring, Maize	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.

Field/control	Type	Options/default	Meaning	Example
		flowering trial		
GeoJSON containing one Polygon or MultiPolygon	File upload	Accepted: geojson, json	Uploads the file required for geojson containing one polygon or multipolygon. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Field notes	Long text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Season notes	Long text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Assign field access	Action button	Depends on the active data and prior selections.	Runs the action named “Assign field access” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

5.2 Tasks, scouting & operations

Path: Fields & Operations > Tasks, scouting & operations

Plan fieldwork, assign responsibilities, capture geolocated observations and photos, and maintain a structured operational diary.

Worked example

Example: assign a task to inspect lodging in Field 2, record a geolocated observation with severity and photo, then log the corrective operation.

Tabs and views

Tab/view	Purpose
Tasks	Opens the tasks stage or result view.
Scouting	Opens the scouting stage or result view.
Operations & inputs	Opens the operations & inputs stage or result view.
Field diary	Opens the field diary stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.

2. Configure the main inputs: Field, Instructions / acceptance criteria, Update a task, Location.
3. Run “Update task status”.
4. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Field	Drop-down	Options: generated from available data	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Instructions / acceptance criteria	Long text	Depends on the active data and prior selections.	Enter the requested text for “Instructions / acceptance criteria”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Update a task	Drop-down	Options: generated from the current data or context	Select one option for “Update a task”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Location	Single choice	Options: Field centroid, Enter GPS coordinates	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Linked task	Drop-down	Options: generated from the current data or context	Select one option for “Linked task”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Observation notes	Long text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Recommended follow-up	Long text	Depends on the active data and prior selections.	Enter the requested text for “Recommended follow-up”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Photo (optional)	File upload	Accepted: jpg, jpeg, png, webp	Uploads the file required for photo (optional). Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Create a follow-up	Checkbox	Depends on the active	Enables or disables	Toggle only when the

Field/control	Type	Options/default	Meaning	Example
scouting task		data and prior selections.	“Create a follow-up scouting task”. Leave it at the default unless the alternative is intentional.	displayed workflow requires it.
Operation notes, weather constraints and safety record	Long text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Update task status	Action button	Depends on the active data and prior selections.	Runs the action named “Update task status” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

5.3 Sensors, irrigation & nutrition

Path: Fields & Operations > Sensors, irrigation & nutrition

Register field sensors, import time series, audit data quality, connect root-zone outputs and record soil, tissue and water samples.

Worked example

Example: register a soil-moisture sensor at 30 cm, import its CSV time series and compare readings with root-zone depletion.

Tabs and views

Tab/view	Purpose
Sensor registry	Opens the sensor registry stage or result view.
Readings	Opens the readings stage or result view.
Quality control	Opens the quality control stage or result view.
Irrigation advisory	Opens the irrigation advisory stage or result view.
Nutrition samples	Opens the nutrition samples stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Field, Source / manufacturer / station ID, Calibration and installation notes, Sensor.
3. Run “Create irrigation task from advisory”.
4. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Field	Drop-down	Options: generated from available data	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Source / manufacturer / station ID	Text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Calibration and installation notes	Long text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Sensor	Drop-down	Options: generated from the current data or context	Select one option for "Sensor". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Upload readings CSV	File upload	Accepted: csv	Uploads the file required for upload readings csv. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Inspect quality flags	Drop-down	Options: generated from the current data or context	Select one option for "Inspect quality flags". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Application efficiency	Slider	Range: 0.3 to 1.0 Default: 0.85	Fraction of gross irrigation that reaches and is stored in the root zone.	0.75
Laboratory method, units, depth and notes	Long text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Import sensor readings	Action button	Depends on the active data and prior selections.	Runs the action named "Import sensor readings" using the current	Review inputs, then run the action and inspect any warnings.

Action	Type	Conditions/default	What it does	Good practice
			selections.	
Create irrigation task from advisory	Action button	Depends on the active data and prior selections.	Runs the action named “Create irrigation task from advisory” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

5.4 Crop health, rules & alerts

Path: Fields & Operations > Crop health, rules & alerts

Combine satellite indices, daily weather, root-zone conditions and sensor data into transparent field-level alerts and attention rankings.

Worked example

Example: create an alert when NDVI falls sharply, NDMI is low and the root-zone stress coefficient Ks is below a selected threshold.

Tabs and views

Tab/view	Purpose
Field health	Opens the field health stage or result view.
Alert rules	Opens the alert rules stage or result view.
Alerts	Opens the alerts stage or result view.
Field leaderboard	Opens the field leaderboard stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Field, Satellite indices, Interpretation and validation note, Update alert.
3. Run “Evaluate enabled alert rules”.
4. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Field	Drop-down	Options: generated from available data	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Satellite indices	Multi-select	Options: available numeric metrics Default: generated from the current data or context	Select one or more entries for “Satellite indices”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.

Field/control	Type	Options/default	Meaning	Example
Interpretation and validation note	Long text	Depends on the active data and prior selections.	Enter the requested text for “Interpretation and validation note”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Update alert	Drop-down	Options: generated from the current data or context	Select one option for “Update alert”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Evaluate enabled alert rules	Action button	Depends on the active data and prior selections.	Runs the action named “Evaluate enabled alert rules” using the current selections.	Review inputs, then run the action and inspect any warnings.
Update alert status	Action button	Depends on the active data and prior selections.	Runs the action named “Update alert status” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

Remote-sensing note

Satellite indices measure spectral response at pixel scale. Small plots, mixed pixels, clouds, shadows and management differences can affect interpretation.

5.5 Management zones & prescriptions

Path: Fields & Operations > Management zones & prescriptions

Build transparent management zones from georeferenced samples, generate field-contained sampling grids and export generic variable-rate exchange files.

Worked example

Example: upload georeferenced soil-organic-matter samples, create three zones and export a generic GeoJSON variable-rate prescription.

Tabs and views

Tab/view	Purpose
Management zones	Opens the management zones stage or result view.
Sampling grid	Opens the sampling grid stage or result view.
Saved prescriptions	Opens the saved prescriptions stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Field, Upload point CSV with latitude, longitude and one or more numeric variables, Rate unit, Prescription name.
3. Run “Generate sampling grid”.
4. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Field	Drop-down	Options: generated from available data	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Upload point CSV with latitude, longitude and one or more numeric variables	File upload	Accepted: csv	Uploads the file required for upload point csv with latitude, longitude and one or more numeric variables. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Rate unit	Text	Default: kg/ha	Enter the requested text for “Rate unit”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Prescription name	Text	Default: {metadata['variable']} management zones	Enter the requested text for “Prescription name”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Build management zones	Action button	Depends on the active data and prior selections.	Runs the action named “Build management zones” using the current selections.	Review inputs, then run the action and inspect any warnings.
Save prescription	Action button	Depends on the active data and prior selections.	Runs the action named “Save prescription” using the current selections.	Review inputs, then run the action and inspect any warnings.
Generate sampling grid	Action button	Depends on the active data and prior selections.	Runs the action named “Generate sampling grid” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

Remote-sensing note

Satellite indices measure spectral response at pixel scale. Small plots, mixed pixels, clouds, shadows and management differences can affect interpretation.

6. AgroLattice Twin

The Twin workspace combines multiple data streams into a current field state, explores alternative scenarios, prioritises measurements and stores calibration settings.

Workspace selector

Open AgroLattice Twin from the sidebar. Use the Workspace view selector and, where present, the Analysis selector to move between related functions without leaving the workspace.

6.1 AgroLattice live twin

Path: AgroLattice Twin > AgroLattice live twin

Unify crop development, weather, root-zone water, Sentinel-2, sensors and field observations into a continuously updated field and plot state.

Worked example

Example: open Field 3 and view predicted male flowering progress, female silking progress, water stress, NDVI and data uncertainty on the same plot map.

Tabs and views

Tab/view	Purpose
Current state	Shows the current field and plot state at the selected date.
Plot intelligence map	Colours plots by selected Twin metrics.
Evidence & provenance	Lists data sources and evidence used by the Twin.
Saved history	Reviews previously saved Twin snapshots.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: AgroLattice Twin, Twin state date, Map metric.
3. Run “Save current twin snapshot”.
4. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
AgroLattice Twin	Drop-down	Options: generated from available data	Select one option for “AgroLattice Twin”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
Twin state date	Date	Default: generated from the current data or context	Sets the date on which the field state is reconstructed.	Latest available date
Map metric	Drop-down	Options: available numeric metrics	Selects the variable used to colour mapped plots in the Twin.	Inspection priority

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Save current twin snapshot	Action button	Depends on the active data and prior selections.	Runs the action named “Save current twin snapshot” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

6.2 AgroLattice scenario studio

Path: AgroLattice Twin > AgroLattice scenario studio

Move through the season and compare transparent what-if scenarios for weather, irrigation, sowing synchrony and planting density.

Worked example

Example: compare the current plan with a scenario that adds 25 mm irrigation and delays male sowing by two days.

Tabs and views

Tab/view	Purpose
Time-travel twin	Opens the time-travel twin stage or result view.
What-if simulator	Opens the what-if simulator stage or result view.
Historical-state comparison	Opens the historical-state comparison stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: AgroLattice Twin, Baseline state date, Move through the season, Scenario name.
3. Run “Save scenario”.
4. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
AgroLattice Twin	Drop-down	Options: generated from available data	Select one option for “AgroLattice Twin”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Baseline state date	Date	Default: generated from the current data or context	Defines the reference period or scenario against which change is measured.	1985-2014
Move through the season	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context	Enter or adjust the numeric setting “Move through the season”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Scenario name	Text	Default: Alternative management scenario	Selects or defines an alternative assumption set for comparison with the baseline.	Add 25 mm irrigation
Comparison metric	Drop-down	Options: generated from the current data or context	Selects the result variable used in a chart, ranking or comparison.	RMSE

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Save scenario	Action button	Depends on the active data and prior selections.	Runs the action named “Save scenario” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

What-if scenarios are exploratory response estimates unless supported by calibrated causal or process-based models.

6.3 AgroLattice adaptive copilot

Path: AgroLattice Twin > AgroLattice adaptive copilot

Prioritise the plots and measurements that reduce uncertainty most, create field tasks, and refine the next replicated trial design.

Worked example

Example: rank plots by uncertainty and create tasks to measure silking percentage and plant height in the plots that would reduce model uncertainty most.

Tabs and views

Tab/view	Purpose
What to measure next	Opens the what to measure next stage or result view.
Next-season design	Opens the next-season design stage or result view.
Create field tasks	Opens the create field tasks stage or result view.
Recommendation register	Opens the recommendation register stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: AgroLattice Twin, Recommendation date, Maximum plots to prioritise, Task due date.
3. Run “Create selected inspection tasks”.
4. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
AgroLattice Twin	Drop-down	Options: generated from available data	Select one option for “AgroLattice Twin”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Recommendation date	Date	Default: generated from the current data or context	Sets the date or time for “Recommendation date”. Confirm that it matches the crop season and local field records.	Use the recorded field date rather than an approximate date when available.
Maximum plots to prioritise	Slider	Range: 3 to 20 Default: 8	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Task due date	Date	Default: generated from the current data or context	Sets the date or time for “Task due date”. Confirm that it matches the crop season and local field records.	Use the recorded field date rather than an approximate date when available.
Assign to	Text	Depends on the active data and prior selections.	Enter the requested text for “Assign to”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Recommendation to close	Drop-down	Options: generated from the current data or context	Select one option for “Recommendation to close”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Save current recommendations	Action button	Depends on the active data and prior selections.	Runs the action named “Save current recommendations” using the current selections.	Review inputs, then run the action and inspect any warnings.
Create selected inspection tasks	Action button	Depends on the active data and prior selections.	Runs the action named “Create selected inspection tasks” using the current selections.	Review inputs, then run the action and inspect any warnings.
Mark resolved	Action button	Depends on the active data and prior selections.	Runs the action named “Mark resolved” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

The copilot prioritises information collection. It does not replace agronomic judgement or authorise changes to the experimental protocol.

6.4 AgroLattice twin configuration

Path: AgroLattice Twin > AgroLattice twin configuration

Link fields and trials, calibrate parent-specific thermal targets, inspect data readiness and export the complete twin record.

Worked example

Example: link a mapped field to a maize trial and replace provisional GDD targets with observed parent-specific median values after the first season.

Tabs and views

Tab/view	Purpose
Create/link twin	Opens the create/link twin stage or result view.
Calibration settings	Opens the calibration settings stage or result view.
Data readiness	Opens the data readiness stage or result view.
Export & model registry	Opens the export & model registry stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: AgroLattice Twin, Mapped field, Maize flowering trial, Twin name.
3. Run “Download complete AgroLattice Twin package”.
4. Review warnings, assumptions, provenance and the main table, chart or map.

5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
AgroLattice Twin	Drop-down	Options: generated from available data	Select one option for “AgroLattice Twin”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Mapped field	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Maize flowering trial	Drop-down	Options: generated from the current data or context	Select one option for “Maize flowering trial”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Twin name	Text	Default: generated from the current data or context	Enter the requested text for “Twin name”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Notes	Long text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Target seed-set percentage	Number	Range: 0.0 to 100.0 Default: generated from the current data or context	Records or predicts the percentage of potential kernel sites that produce filled kernels.	88%
Allow clearly labelled heuristic fallback when calibrated models are unavailable	Checkbox	Default: generated from the current data or context	Enables or disables “Allow clearly labelled heuristic fallback when calibrated models are unavailable”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download complete AgroLattice Twin package	Download	Depends on the active data and prior selections.	Downloads the current complete agrolattice twin package for archiving,	Save it with the project, country and analysis date in the filename.

Action	Type	Conditions/default	What it does	Good practice
			analysis or reporting.	

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

7. Climate & Earth Observation

This workspace analyses historical climate, similarity, spatial patterns, hazards and Sentinel-2 observations. It is used for location comparison, climate-risk screening and remote monitoring.

Workspace selector

Open Climate & Earth Observation from the sidebar. Use the Workspace view selector and, where present, the Analysis selector to move between related functions without leaving the workspace.

7.1 Climate similarity

Path: Climate & Earth Observation > Climate similarity

Provides the climate similarity function within AgroLattice.

Worked example

Example: use Montecillo in 2025 as the reference and find the most similar locations using maize-focused temperature, rainfall and radiation weights.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Select year, Select plant for default weights, Select reference city, Show raw cosine similarity (-1 to 1).
3. Run “Calculate Area Similarities”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Select year	Drop-down	Options: available years	Selects a calendar or season year included in the analysis.	2025
Select plant for default weights	Drop-down	Options: General, Maize, Coffee	Controls the relative influence of a variable or component in a combined score.	Temperature weight = 1.5
Select reference city	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Show raw cosine similarity (-1 to 1)	Checkbox	Default: False	Enables or disables “Show raw cosine similarity (-1 to 1)”. Leave it at the default unless the alternative is intentional. Terms: RAW: Readily available water: the fraction of TAW that can be depleted before crop water stress is expected.	Toggle only when the displayed workflow requires it.
Use disk cache	Checkbox	Default: False	Enables or disables “Use disk cache”. Leave it at	Toggle only when the displayed workflow

Field/control	Type	Options/default	Meaning	Example
			the default unless the alternative is intentional.	requires it.
Show similarity heatmap overlay	Checkbox	Default: False	Enables or disables “Show similarity heatmap overlay”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Clear similarity cache	Action button	Depends on the active data and prior selections.	Performs a destructive or clearing action. Review confirmation controls and backups before using it. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Use only after verifying the active country, project, field or trial.
Calculate Area Similarities	Action button	Depends on the active data and prior selections.	Runs the action named “Calculate Area Similarities” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download Similarity Results CSV	Download	Depends on the active data and prior selections.	Downloads the current similarity results csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Generate Similarity Map	Action button	Depends on the active data and prior selections.	Runs the action named “Generate Similarity Map” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

A high score means the weighted monthly climate vectors are close; it does not prove that soils, pests, management or crop performance will also be similar.

7.2 Compare two cities

Path: Climate & Earth Observation > Compare two cities

Compare two cities across climate profiles, trends, water stress, crop reliability, and climate-limited yield.

Worked example

Example: compare Nicosia and a selected Mexican city for temperature, rainfall, trends, water stress and crop suitability.

Tabs and views

Tab/view	Purpose
Overview	Opens the overview stage or result view.
Monthly climate	Opens the monthly climate stage or result view.
Trends	Opens the trends stage or result view.
Crop and water	Opens the crop and water stage or result view.
Downloads	Opens the downloads stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Crop profile, Crop stage / window, City A, City B.
3. Run “Download interpretation as Markdown”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Crop profile	Drop-down	Options: generated from the current data or context	Select one option for “Crop profile”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Crop stage / window	Drop-down	Options: generated from the current data or context	Selects a crop-development phase such as establishment, flowering or maturity.	Flowering
City A	Drop-down	Options: generated from the current data or context	Select one option for “City A”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
City B	Drop-down	Options: generated from the current data or context	Select one option for “City B”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			reduce some background and atmospheric effects.	
Climate comparison period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Climate months included	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Climate months included”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Climate variables	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Climate variables”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Months used for crop analysis	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Months used for crop analysis”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Yield-screening year	Drop-down	Options: available years	Selects a calendar or season year included in the analysis.	2025
Variable shown in the monthly profile	Drop-down	Options: generated from the current data or context	Select one option for “Variable shown in the monthly profile”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Trend variable to compare	Drop-down	Options: generated from the current data or context	Select one option for “Trend variable to compare”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download Pairwise HTML Report	Download	Depends on the active data and prior selections.	Downloads the current pairwise html report for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download All Pairwise Tables (ZIP)	Download	Depends on the active data and prior selections.	Downloads the current all pairwise tables (zip) for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

7.3 Climate analogues

Path: Climate & Earth Observation > Climate analogues

Provides the climate analogues function within AgroLattice.

Worked example

Example: identify historical years or locations whose climate resembles a future or recent target condition.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Select reference city, Reference climate year, Comparison city-pool year, Climate variables used for analogue matching.
3. Run “Generate Climate Analogue Map”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Select reference city	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Reference climate year	Drop-down	Options: available years	Selects a calendar or season year included in the analysis.	2025
Comparison city-pool year	Drop-down	Options: available years	Selects a calendar or season year included in the analysis.	2025
Climate variables used for analogue matching	Multi-select	Options: generated from the current data or context Default:	Select one or more entries for “Climate variables used for analogue	Select only variables, locations or treatments required by the analysis.

Field/control	Type	Options/default	Meaning	Example
		generated from the current data or context	matching”. Selecting more entries broadens the analysis and may increase processing time.	
Number of top analogues to display	Slider	Range: 5 to 50 Default: 15	Enter or adjust the numeric setting “Number of top analogues to display”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Show raw analogue similarity (-1 to 1)	Checkbox	Default: False	Enables or disables “Show raw analogue similarity (-1 to 1)”. Leave it at the default unless the alternative is intentional. Terms: RAW: Readily available water; the fraction of TAW that can be depleted before crop water stress is expected.	Toggle only when the displayed workflow requires it.
Show analogue heatmap overlay	Checkbox	Default: False	Enables or disables “Show analogue heatmap overlay”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download Analogue Results CSV	Download	Depends on the active data and prior selections.	Downloads the current analogue results csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Generate Climate Analogue Map	Action button	Depends on the active data and prior selections.	Runs the action named “Generate Climate Analogue Map” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

7.4 Explain similarity

Path: Climate & Earth Observation > Explain similarity

Open the similarity score and show which variables and months make two locations alike or different.

Worked example

Example: explain why two locations received a high similarity score and identify the months and variables that create the remaining differences.

Tabs and views

Tab/view	Purpose
Similarity drivers	Opens the similarity drivers stage or result view.
Difference drivers	Opens the difference drivers stage or result view.
Monthly heatmap	Opens the monthly heatmap stage or result view.
Feature detail	Opens the feature detail stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Analysis year, Weight profile, Reference city, Comparison city.
3. Run “Download interpretation as Markdown”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Analysis year	Drop-down	Options: available years	Selects a calendar or season year included in the analysis.	2025
Weight profile	Drop-down	Options: General, Maize, Coffee	Controls the relative influence of a variable or component in a combined score.	Temperature weight = 1.5
Reference city	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Comparison city	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download Explanation CSV	Download	Depends on the active data and prior selections.	Downloads the current explanation csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

7.5 Weight sensitivity

Path: Climate & Earth Observation > Weight sensitivity

Test whether similarity rankings remain stable when the model weights are changed across many plausible scenarios.

Worked example

Example: perturb similarity weights across many scenarios and check whether the same top five locations remain stable.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Analysis year, Baseline weight profile, Reference city, Monte Carlo iterations.
3. Run “Download interpretation as Markdown”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Analysis year	Drop-down	Options: available years	Selects a calendar or season year included in the analysis.	2025
Baseline weight profile	Drop-down	Options: General, Maize, Coffee	Controls the relative influence of a variable or component in a combined score.	Temperature weight = 1.5
Reference city	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Monte Carlo iterations	Slider	Range: 25 to 250 Default: 100	Enter or adjust the numeric setting “Monte Carlo iterations”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Weight variation (±%)	Slider	Range: 10 to 100 Default: 40	Controls the relative influence of a variable or component in a combined score.	Temperature weight = 1.5
Top-k threshold	Slider	Range: 3 to 30 Default: 10	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Random seed	Number	Range: 0 to 1000000 Default: 42	Fixed value used to reproduce a randomisation, clustering or model-fitting process.	2027
Inspect variable sensitivity for one location	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download Summary CSV	Download	Depends on the active data and prior selections.	Downloads the current summary csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download Simulation Ranks CSV	Download	Depends on the active data and prior selections.	Downloads the current simulation ranks csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download Sampled Weights CSV	Download	Depends on the active data and prior selections.	Downloads the current sampled weights csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

7.6 Agroclimatic clusters

Path: Climate & Earth Observation > Agroclimatic clusters

Provides the agroclimatic clusters function within AgroLattice.

Worked example

Example: use PCA and K-means to group Cyprus locations into meaningful climate zones while allowing duplicate NASA grid profiles to collapse safely.

Tabs and views

Tab/view	Purpose
PCA map	Opens the pca map stage or result view.
Cluster profiles	Opens the cluster profiles stage or result view.
Model diagnostics	Opens the model diagnostics stage or result view.
Geographic map	Opens the geographic map stage or result view.
Tables	Opens the tables stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Clustering period, Months included, Climate variables, Number of clusters.
3. Run “Generate Agroclimatic Cluster Map”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Clustering period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Months included	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Months included”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Climate variables	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Climate variables”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Number of clusters	Single choice	Options: Choose automatically, Set manually	Sets how many groups are requested. Use no more than the number of distinct usable climate profiles.	3 clusters
Requested clusters	Slider	Range: 2 to 10 Default: 5	Sets how many groups are requested. Use no more than the number of distinct usable climate profiles.	3 clusters
Minimum location data completeness	Slider	Range: 0.5 to 1.0 Default: 0.8	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Random seed	Number	Range: 0 to 1000000 Default: 42	Fixed value used to reproduce a randomisation, clustering or model-fitting process.	2027

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Generate Agroclimatic Cluster Map	Action button	Depends on the active data and prior selections.	Runs the action named “Generate Agroclimatic Cluster Map” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download Cluster Assignments CSV	Download	Depends on the active data and prior selections.	Downloads the current cluster assignments csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download PCA Feature	Download	Depends on the active	Downloads the current	Save it with the project,

Action	Type	Conditions/default	What it does	Good practice
Loadings CSV		data and prior selections.	pca feature loadings csv for archiving, analysis or reporting. Terms: PCA: Principal component analysis: a dimension-reduction method that summarises correlated climate variables into orthogonal axes.	country and analysis date in the filename.

Interpreting the result

Clusters are descriptive groupings. PCA axes are mathematical combinations of variables and should be interpreted using loadings, not named automatically as biological processes.

7.7 Spatial intelligence

Path: Climate & Earth Observation > Spatial intelligence

Explore climate-space geometry, migration trajectories, geographically constrained analogues, state summaries, and optional choropleth maps.

Worked example

Example: trace how a location moves through climate space over time and find geographically constrained analogues within a chosen radius.

Tabs and views

Tab/view	Purpose
Climate space	Opens the climate space stage or result view.
Migration trajectories	Opens the migration trajectories stage or result view.
Constrained similarity	Opens the constrained similarity stage or result view.
State summaries & choropleths	Opens the state summaries & choropleths stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Period, Months, Variables, Embedding method.
3. Run “Generate filled state choropleth”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Period	Slider	Range: generated from the current data or context to generated from the current data or context Default:	Defines the years or dates included in the calculation.	1991-2025

Field/control	Type	Options/default	Meaning	Example
		generated from the current data or context, generated from the current data or context		
Months	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Selects the months included in a crop window or climate summary.	May-September
Variables	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Variables”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Embedding method	Drop-down	Options: PCA, UMAP, t-SNE	Select one option for “Embedding method”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Minimum location completeness	Slider	Range: 0.5 to 1.0 Default: 0.8	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Colour by K-means clusters	Slider	Range: 2 to 10 Default: 5	Sets how many groups are requested. Use no more than the number of distinct usable climate profiles.	3 clusters
Reference city	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Reference year	Drop-down	Options: available years	Selects a calendar or season year included in the analysis.	2025
Comparison-year interval	Slider	Range: 1 to 10 Default: 5	Enter or adjust the numeric setting “Comparison-year interval”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Exclude the reference location from candidate analogues	Checkbox	Default: False	Enables or disables “Exclude the reference location from candidate analogues”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
Optional location metadata CSV (CITY, STATE, Elevation_m, Landscape)	File upload	Accepted: csv	Uploads the file required for optional location metadata csv (city, state, elevation_m, landscape). Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.

Field/control	Type	Options/default	Meaning	Example
Year	Drop-down	Options: available years	Selects a calendar or season year included in the analysis.	2025
Distance range (km)	Slider	Range: 0 to 3000 Default: 0, 1500	Enter or adjust the numeric setting “Distance range (km)”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
State constraint	Drop-down	Options: All states, Same state only, Selected states	Select one option for “State constraint”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Allowed states	Multi-select	Options: generated from the current data or context	Select one or more entries for “Allowed states”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Apply elevation tolerance when metadata are available	Checkbox	Default: False	Enables or disables “Apply elevation tolerance when metadata are available”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
Elevation tolerance (m)	Number	Range: 0.0 to 4000.0 Default: 500.0	Enter or adjust the numeric setting “Elevation tolerance (m)”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Landscape category	Drop-down	Options: generated from the current data or context	Select one option for “Landscape category”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Variable	Drop-down	Options: generated from the current data or context	Select one option for “Variable”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			atmospheric effects.	
Summary statistic	Drop-down	Options: Mean, Median, Maximum, Minimum	Select one option for “Summary statistic”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Optional {selected_country} administrative-area GeoJSON for a filled choropleth	File upload	Accepted: geojson, json	Uploads the file required for optional {selected_country} administrative-area geojson for a filled choropleth. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
GeoJSON property containing state names	Drop-down	Options: generated from the current data or context	Select one option for “GeoJSON property containing state names”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download climate-space coordinates	Download	Depends on the active data and prior selections.	Downloads the current climate-space coordinates for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download constrained-similarity results	Download	Depends on the active data and prior selections.	Downloads the current constrained-similarity results for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download location-metadata template	Download	Depends on the active data and prior selections.	Downloads the current location-metadata template for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download state summary	Download	Depends on the active data and prior selections.	Downloads the current state summary for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Action	Type	Conditions/default	What it does	Good practice
Generate filled state choropleth	Action button	Depends on the active data and prior selections.	Runs the action named “Generate filled state choropleth” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

7.8 Composite agricultural risk

Path: Climate & Earth Observation > Composite agricultural risk

Combine water stress, heat exposure, rainfall variability, warming, anomalies, and optional crop instability while retaining every component for interpretation.

Worked example

Example: combine drought, heat, rainfall variability, warming and crop instability into a transparent screening score while retaining every component.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Analysis period, Months included, Include crop-instability risk, Crop.
3. Run “Generate composite-risk map”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Analysis period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Months included	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Months included”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Include crop-instability risk	Checkbox	Default: True	Enables or disables “Include crop-instability risk”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
Crop	Drop-down	Options: generated from the current data or	Select the crop whose validated parameters or	Maize

Field/control	Type	Options/default	Meaning	Example
		context	custom profile should be used.	
Crop stage	Drop-down	Options: generated from the current data or context	Selects a crop-development phase such as establishment, flowering or maturity.	Flowering
Inspect one location	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download composite-risk CSV	Download	Depends on the active data and prior selections.	Downloads the current composite-risk csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Generate composite-risk map	Action button	Depends on the active data and prior selections.	Runs the action named “Generate composite-risk map” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

7.9 Trend analysis

Path: Climate & Earth Observation > Trend analysis

Quantify monotonic climate change using Mann–Kendall tests, Sen's slope, confidence intervals, and linear trends.

Worked example

Example: estimate the Sen slope for annual maximum temperature and test whether the monotonic trend is statistically supported.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Trend-analysis level, Climate variable, Trend-analysis period, Months included.
3. Run “Generate Trend Map”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Trend-analysis level	Single choice	Options: Cities, States	Select one option for “Trend-analysis level”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Climate variable	Drop-down	Options: generated from the current data or context	Select one option for “Climate variable”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Trend-analysis period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Months included	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Months included”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Significance level	Selection slider	Default: 0.05	Enter or adjust the numeric setting “Significance level”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Select city for detailed trend plot	Drop-down	Options: generated from the current data or context	Select one option for “Select city for detailed trend plot”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Select state for detailed trend plot	Drop-down	Options: generated from the current data or context	Select one option for “Select state for detailed trend plot”. Available choices may depend on the active country,	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download Trend Results CSV	Download	Depends on the active data and prior selections.	Downloads the current trend results csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Generate Trend Map	Action button	Depends on the active data and prior selections.	Runs the action named "Generate Trend Map" using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Sen slope describes the typical rate of change per year. A statistically significant trend is not automatically agronomically important.

7.10 Climate anomalies

Path: Climate & Earth Observation > Climate anomalies

Compare a recent or selected period with a historical baseline using absolute, percentage, and standardised anomalies.

Worked example

Example: compare 2016-2025 temperature and rainfall with a 1985-2014 baseline using absolute and standardised anomalies.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Anomaly-analysis level, Climate variable, Baseline period, Comparison period.
3. Run "Generate Anomaly Map".
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Anomaly-analysis level	Single choice	Options: Cities, States	Select one option for “Anomaly-analysis level”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Climate variable	Drop-down	Options: generated from the current data or context	Select one option for “Climate variable”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Baseline period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Comparison period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Months included	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Months included”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Select city for detailed plot	Drop-down	Options: generated from the current data or context	Select one option for “Select city for detailed plot”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Select state for detailed plot	Drop-down	Options: generated from the current data or context	Select one option for “Select state for detailed plot”. Available choices	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	
Anomaly map metric	Drop-down	Options: Absolute Anomaly, % Anomaly, Standardized Anomaly (z)	Selects the result variable used in a chart, ranking or comparison.	RMSE

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download Anomaly Results CSV	Download	Depends on the active data and prior selections.	Downloads the current anomaly results csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Generate Anomaly Map	Action button	Depends on the active data and prior selections.	Runs the action named “Generate Anomaly Map” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

A positive anomaly means the target period is above the baseline for that variable. Standardised anomalies express the difference in baseline standard deviations.

7.11 Drought & water stress

Path: Climate & Earth Observation > Drought & water stress

Measure whether precipitation meets atmospheric water demand and identify locations with persistent water-balance pressure.

Worked example

Example: rank locations by persistent precipitation minus evapotranspiration deficit during the crop season.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Water-stress analysis level, Analysis period, Months included, Select city for detailed water-stress plot.
3. Run “Generate Water-Stress Map”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Water-stress analysis level	Single choice	Options: Cities, States	Select one option for “Water-stress analysis level”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Analysis period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Months included	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Months included”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Select city for detailed water-stress plot	Drop-down	Options: generated from the current data or context	Select one option for “Select city for detailed water-stress plot”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Select state for detailed water-stress plot	Drop-down	Options: generated from the current data or context	Select one option for “Select state for detailed water-stress plot”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Water-stress map metric	Drop-down	Options: Water Stress Score, Water Deficit P - ETo (mm/day), Estimated Annual Water Deficit (mm/year), Aridity Index P / ETo, Dryness Ratio ETo / P	Selects the result variable used in a chart, ranking or comparison.	RMSE

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download Water-Stress Results CSV	Download	Depends on the active data and prior selections.	Downloads the current water-stress results csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Generate Water-Stress Map	Action button	Depends on the active data and prior selections.	Runs the action named “Generate Water-Stress Map” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Negative climatic water balance indicates atmospheric demand exceeds precipitation; it is a screening measure and does not replace measured soil moisture.

7.12 Climate hazards laboratory

Path: Climate & Earth Observation > Climate hazards laboratory

Analyse monthly extremes, structural change points, interannual variability, seasonal drought signals, and uploaded future-climate scenarios.

Worked example

Example: identify heat extremes, rainfall change points, high variability and seasonal drought signals, then apply an uploaded future scenario.

Tabs and views

Tab/view	Purpose
Monthly extremes	Opens the monthly extremes stage or result view.
Change points	Opens the change points stage or result view.
Climate variability	Opens the climate variability stage or result view.
Seasonal drought indices	Opens the seasonal drought indices stage or result view.
Future scenarios	Opens the future scenarios stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Location, Baseline period, Analysis period, Hot-month percentile.
3. Run “Download interpretation as Markdown”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Location	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Baseline period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Analysis period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Hot-month percentile	Slider	Range: 0.8 to 0.99 Default: 0.95	Enter or adjust the numeric setting “Hot-month percentile”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Dry-month percentile	Slider	Range: 0.01 to 0.3 Default: 0.1	Enter or adjust the numeric setting “Dry-month percentile”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Wet-month percentile	Slider	Range: 0.7 to 0.99 Default: 0.9	Enter or adjust the numeric setting “Wet-month percentile”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Analysis level	Single choice	Options: Cities, States	Select one option for “Analysis level”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
State	Drop-down	Options: generated from the current data or context	Select one option for “State”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			canopy sensitivity and reduce some background and atmospheric effects.	
Variable	Drop-down	Options: generated from the current data or context	Select one option for “Variable”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Months	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Selects the months included in a crop window or climate summary.	May-September
Rolling variability window (years)	Slider	Range: 2 to 10 Default: 5	Enter or adjust the numeric setting “Rolling variability window (years)”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Inspect rolling variability	Drop-down	Options: generated from the current data or context	Select one option for “Inspect rolling variability”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Accumulation window (months)	Selection slider	Range: 1, 3, 6, 12 to context dependent Default: 3	Enter or adjust the numeric setting “Accumulation window (months)”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Future climate scenario CSV	File upload	Accepted: csv	Uploads the file required for future climate scenario csv. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Scenario	Drop-down	Options: generated from	Selects or defines an	Add 25 mm irrigation

Field/control	Type	Options/default	Meaning	Example
		the current data or context	alternative assumption set for comparison with the baseline.	
Historical baseline	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the reference period or scenario against which change is measured.	1985-2014
Future period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Also compare crop reliability	Checkbox	Default: True	Enables or disables “Also compare crop reliability”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
Crop	Drop-down	Options: generated from the current data or context	Select the crop whose validated parameters or custom profile should be used.	Maize
Crop stage	Drop-down	Options: generated from the current data or context	Selects a crop-development phase such as establishment, flowering or maturity.	Flowering

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download future-scenario template	Download	Depends on the active data and prior selections.	Downloads the current future-scenario template for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download scenario comparison	Download	Depends on the active data and prior selections.	Downloads the current scenario comparison for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

7.13 Satellite crop monitoring

Path: Climate & Earth Observation > Satellite crop monitoring

Retrieve Sentinel-2 Level-2A imagery for a field or point buffer, apply pixel-level cloud masking, quantify vegetation and canopy-moisture indices, compare seasons, and test agreement with the phenology and root-zone models.

Worked example

Example: draw a field boundary, search Sentinel-2 scenes, process monthly low-cloud images and inspect NDVI, NDMI and NDRE trends.

Tabs and views

Tab/view	Purpose
Search & time series	Find scenes, process selected imagery and build index time series.
Image explorer	Preview true-colour imagery and selected vegetation indices.
Historical anomalies	Compare one season with baseline seasons.
Model integration	Join satellite results with phenology and root-zone outputs.
Cache & methods	Inspect data provenance, dependencies, cache and methods.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Area source, City, Buffer radius (m), Upload a Polygon or MultiPolygon GeoJSON.
3. Run “Search Sentinel-2 catalogue”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Area source	Single choice	Options: generated from the current data or context	Choose where the satellite analysis geometry comes from: a city buffer, uploaded polygon, current project, mapped field, trial plot or current session field.	Mapped field
City	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Buffer radius (m)	Slider	Range: 100 to 5000 Default: 500	Sets the radius around a selected point when a polygon field boundary is unavailable.	500 m
Upload a Polygon or MultiPolygon GeoJSON	File upload	Accepted: geojson, json	Uploads the file required for upload a polygon or multipolygon geojson. Check the displayed	Use the matching template supplied by AgroLattice.

Field/control	Type	Options/default	Meaning	Example
			template or required column names before import.	
Field or analysis-area name	Text	Default: generated from the current data or context	Enter the requested text for “Field or analysis-area name”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Crop	Drop-down	Options: available crops	Select the crop whose validated parameters or custom profile should be used.	Maize
Planting or season start	Date	Default: generated from the current data or context	Sets the date or time for “Planting or season start”. Confirm that it matches the crop season and local field records.	Use the recorded field date rather than an approximate date when available.
Search start	Date	Default: generated from the current data or context	Sets the date or time for “Search start”. Confirm that it matches the crop season and local field records.	Use the recorded field date rather than an approximate date when available.
Search end	Date	Default: generated from the current data or context	Sets the date or time for “Search end”. Confirm that it matches the crop season and local field records.	Use the recorded field date rather than an approximate date when available.
Maximum scene cloud (%)	Slider	Range: 0 to 100 Default: 50	Filters satellite catalogue scenes using the scene-level cloud metadata before pixel-level masking.	30%
Maximum catalogue items	Number	Range: 10 to 1000 Default: 300	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Catalogue provider	Drop-down	Options: Automatic failover, Earth Search, Planetary Computer Built-in help: Automatic tries Earth Search first and switches to Planetary Computer when Earth Search is unavailable.	Select the STAC catalogue used to find Sentinel-2 scenes. Automatic failover tries another provider when the first is unavailable.	Automatic failover
Indices	Multi-select	Options: generated from the current data or context Default: NDVI, EVI, NDMI, NDRE	Select vegetation and moisture indices to calculate from each accepted scene.	NDVI, EVI, NDMI and NDRE
Scene-selection rule	Drop-down	Options: Lowest-cloud scene per month, Evenly sample maximum N, Lowest-cloud maximum N, All search results	Chooses how catalogue scenes are reduced before processing.	Lowest-cloud scene per month
Maximum scenes to process	Number	Range: 1 to 300 Default: 36	Caps the number of scenes processed in one run.	24 scenes
Analysis resolution	Drop-down	Options: 10, 20, 30, 60	Sets the output pixel size. Ten metres preserves the finest Sentinel-2 bands but may increase	10 m

Field/control	Type	Options/default	Meaning	Example
			processing cost.	
Minimum clear field pixels (%)	Slider	Range: 1 to 100 Default: 30	Rejects processed scenes when too little of the field remains after cloud and quality masking.	60%
Include SCL water pixels	Checkbox	Default: False	Enables or disables “Include SCL water pixels”. Leave it at the default unless the alternative is intentional. Terms: SCL: Sentinel-2 Scene Classification Layer, used to mask clouds, shadows, snow and other unwanted pixel classes.	Toggle only when the displayed workflow requires it.
Ignore processed-scene cache	Checkbox	Default: False	Enables or disables “Ignore processed-scene cache”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
Maximum raster pixels per scene	Number	Range: 50000 to 5000000 Default: 2000000	Safety limit that prevents unexpectedly large satellite processing jobs.	500,000 pixels
Scene	Drop-down	Options: generated from the current data or context	Select one option for “Scene”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Index preview	Drop-down	Options: generated from the current data or context	Select one option for “Index preview”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Preview resolution	Drop-down	Options: 10, 20, 30, 60	Controls spatial or temporal detail and computational cost. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	10 m satellite pixels
Index	Drop-down	Options: generated from the current data or context	Select one option for “Index”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	
Target season year	Drop-down	Options: generated from the current data or context	The season to compare against historical years or a baseline.	2025
Season length (days)	Number	Range: 30 to 365 Default: 180	Number of days included after the planting or season-start date.	150 days
Phenology-time bin	Drop-down	Options: 5, 10, 15, 20, 30	Select one option for “Phenology-time bin”. Available choices may depend on the active country, dataset, project or previous selection. Terms: Phenology: The timing of crop development stages such as emergence, flowering and maturity. EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Baseline years	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Defines the reference period or scenario against which change is measured.	1985-2014
Root-zone metric for scatter plot	Drop-down	Options: generated from the current data or context	Selects the result variable used in a chart, ranking or comparison.	RMSE
Confirm clearing the processed satellite cache	Checkbox	Depends on the active data and prior selections.	Enables or disables “Confirm clearing the processed satellite cache”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download active field GeoJSON	Download	Depends on the active data and prior selections.	Downloads the current active field geojson for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Search Sentinel-2 catalogue	Action button	Depends on the active data and prior selections.	Runs the action named “Search Sentinel-2 catalogue” using the current selections.	Review inputs, then run the action and inspect any warnings.

Action	Type	Conditions/default	What it does	Good practice
Download scene catalogue	Download	Depends on the active data and prior selections.	Downloads the current scene catalogue for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Process selected satellite scenes	Action button	Depends on the active data and prior selections.	Runs the action named "Process selected satellite scenes" using the current selections.	Review inputs, then run the action and inspect any warnings.
Download satellite time series	Download	Depends on the active data and prior selections.	Downloads the current satellite time series for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download complete satellite package	Download	Depends on the active data and prior selections.	Downloads the current complete satellite package for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Generate field preview	Action button	Depends on the active data and prior selections.	Runs the action named "Generate field preview" using the current selections. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Review inputs, then run the action and inspect any warnings.
Download true-colour PNG	Download	Depends on the active data and prior selections.	Downloads the current true-colour png for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download index PNG	Download	Depends on the active data and prior selections.	Downloads the current index png for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Calculate historical satellite anomalies	Action button	Depends on the active data and prior selections.	Runs the action named "Calculate historical satellite anomalies" using the current selections.	Review inputs, then run the action and inspect any warnings.
Download historical satellite anomalies	Download	Depends on the active data and prior selections.	Downloads the current historical satellite anomalies for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download stage-level satellite summary	Download	Depends on the active data and prior selections.	Downloads the current stage-level satellite summary for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Compare satellite and root-zone results	Action button	Depends on the active data and prior selections.	Runs the action named "Compare satellite and root-zone results" using the current selections.	Review inputs, then run the action and inspect any warnings.
Clear satellite cache	Action button	Depends on the active data and prior selections.	Performs a destructive or clearing action. Review confirmation controls and backups before using it. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Use only after verifying the active country, project, field or trial.
Download example field	Download	Depends on the active	Downloads the current	Save it with the project,

Action	Type	Conditions/default	What it does	Good practice
GeoJSON		data and prior selections.	example field geojson for archiving, analysis or reporting.	country and analysis date in the filename.

Interpreting the result

Vegetation indices respond to canopy properties but are not direct diagnoses of disease, nutrient deficiency or yield.

Remote-sensing note

Satellite indices measure spectral response at pixel scale. Small plots, mixed pixels, clouds, shadows and management differences can affect interpretation.

8. Crop Decisions

This workspace turns weather and crop parameters into phenology, suitability, soil-water, yield, economics and process-based simulation outputs.

Workspace selector

Open Crop Decisions from the sidebar. Use the Workspace view selector and, where present, the Analysis selector to move between related functions without leaving the workspace.

8.1 Daily weather & phenology

Path: Crop Decisions > Daily weather & phenology

Retrieve NASA POWER daily weather, calculate growing degree days, predict crop-stage timing, quantify stage-specific exposure, and compare a season with its historical distribution.

Worked example

Example: fetch daily weather for Montecillo from 1 May to 30 September, calculate maize GDD and estimate dates for emergence, flowering and maturity.

Tabs and views

Tab/view	Purpose
Daily weather	Opens the daily weather stage or result view.
Phenology & stage exposure	Opens the phenology & stage exposure stage or result view.
Historical context	Opens the historical context stage or result view.
Cache & methods	Inspect data provenance, dependencies, cache and methods.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Heat-day threshold, Tmax (°C), Severe-heat threshold, Tmax (°C), Frost/cold threshold, Tmin (°C), Dry-day precipitation threshold (mm).
3. Run “Run phenology simulation”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Heat-day threshold, Tmax (°C)	Number	Range: -10.0 to 60.0 Default: 35.0	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Severe-heat threshold, Tmax (°C)	Number	Range: -10.0 to 65.0 Default: 40.0	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Frost/cold threshold, Tmin (°C)	Number	Range: -20.0 to 20.0 Default: 0.0	Sets the value at which a condition, alert or classification changes.	Ks below 0.6

Field/control	Type	Options/default	Meaning	Example
Dry-day precipitation threshold (mm)	Number	Range: 0.0 to 20.0 Default: 1.0	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Heavy-rain threshold (mm day ⁻¹)	Number	Range: 1.0 to 300.0 Default: 25.0	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
High-VPD threshold (kPa)	Number	Range: 0.1 to 10.0 Default: 2.0	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
GDD base temperature (°C)	Number	Range: -5.0 to 30.0 Default: 10.0	Controls or selects a temperature threshold, range or variable in degrees Celsius. Terms: GDD: Growing degree days: accumulated thermal time above a base temperature, usually capped at an upper temperature.	30 °C
Apply an upper temperature cap	Checkbox	Default: True	Controls or selects a temperature threshold, range or variable in degrees Celsius.	30 °C
GDD upper cap (°C)	Number	Range: 10.0 to 50.0 Default: 30.0	Enter or adjust the numeric setting “GDD upper cap (°C)”. Use documented units and a scientifically justified value. Terms: GDD: Growing degree days: accumulated thermal time above a base temperature, usually capped at an upper temperature.	Start with the default, then test sensitivity where appropriate.
GDD calculation	Single choice	Options: Modified average, Simple average Built-in help: Modified average raises Tmin to the base and caps Tmax at the selected upper threshold before averaging. Simple average uses mean temperature minus the base.	Select one option for “GDD calculation”. Available choices may depend on the active country, dataset, project or previous selection. Terms: GDD: Growing degree days: accumulated thermal time above a base temperature, usually capped at an upper temperature. EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Location source	Single choice	Options: App city, Custom coordinates	Choose a location from the country catalogue or enter exact custom coordinates.	App city
City	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Latitude	Number	Range: -90.0 to 90.0	North-south coordinate in	19.46

Field/control	Type	Options/default	Meaning	Example
		Default: generated from the current data or context	decimal degrees. Positive values are north of the equator.	
Longitude	Number	Range: -180.0 to 180.0 Default: generated from the current data or context	East-west coordinate in decimal degrees. Mexico and Cyprus longitudes differ in sign; verify the map before downloading data.	-98.90
Start date	Date	Default: generated from the current data or context	Defines the temporal boundary of the analysis or data request.	1 May 2027
End date	Date	Default: generated from the current data or context	Defines the temporal boundary of the analysis or data request.	1 May 2027
NASA POWER parameters	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “NASA POWER parameters”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Time standard	Drop-down	Options: LST, UTC	Choose local solar time (LST) or UTC for NASA POWER daily output.	LST
Ignore local cache and download again	Checkbox	Default: False	Forces a fresh daily-weather request rather than using cached data.	Disabled
Crop	Drop-down	Options: generated from the current data or context	Select the crop whose validated parameters or custom profile should be used.	Maize
Planting or simulation start date	Date	Default: generated from the current data or context	Sets the starting date from which stage durations or GDD are accumulated. Terms: GDD: Growing degree days: accumulated thermal time above a base temperature, usually capped at an upper temperature.	1 May 2027
Stage-prediction method	Single choice	Options: Validated stage-duration ranges, User-defined cumulative GDD targets Built-in help: Duration mode uses the FAO or source-derived stage-duration ranges stored in the validated library. Thermal-time mode requires genotype-specific cumulative GDD targets supplied by the user.	Choose calendar-duration ranges or user-supplied cumulative GDD targets. Terms: GDD: Growing degree days: accumulated thermal time above a base temperature, usually capped at an upper temperature.	User-defined cumulative GDD targets
Stage-duration profile	Drop-down	Options: generated from the current data or context	Selects a crop-development phase such as establishment, flowering or maturity.	Flowering
Duration within each published range	Single choice	Options: Minimum, Midpoint, Maximum	Select one option for “Duration within each published range”.	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	
Maximum simulation length (days)	Number	Range: 30 to 730 Default: 240	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Redownload daily weather instead of using cache	Checkbox	Default: False	Enables or disables “Redownload daily weather instead of using cache”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
Stage-exposure chart	Drop-down	Options: generated from the current data or context	Selects a crop-development phase such as establishment, flowering or maturity.	Flowering
First historical year	Number	Range: 1981 to generated from the current data or context Default: 1991	First year included in the historical comparison.	1991
Last historical year	Number	Range: 1981 to generated from the current data or context Default: generated from the current data or context	Last year included in the historical comparison.	2025
Target season year	Number	Range: 1981 to generated from the current data or context Default: generated from the current data or context	The season to compare against historical years or a baseline.	2025
Season length (days)	Number	Range: 30 to 730 Default: 150	Number of days included after the planting or season-start date.	150 days
Planting month	Drop-down	Options: generated from the current data or context	Selects the months included in a crop window or climate summary.	May-September
Planting day	Number	Range: 1 to 28 Default: 1	Enter or adjust the numeric setting “Planting day”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Minimum season completeness	Slider	Range: 50 to 100 Default: 90	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Redownload historical daily weather	Checkbox	Default: False	Enables or disables “Redownload historical daily weather”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
Historical metric	Drop-down	Options: generated from the current data or context	Selects the result variable used in a chart, ranking or comparison.	RMSE

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Fetch daily weather	Action button	Depends on the active data and prior selections.	Runs the action named “Fetch daily weather” using the current selections. Terms: ETc: Crop evapotranspiration: estimated crop water use, commonly calculated as crop coefficient Kc multiplied by ET0.	Review inputs, then run the action and inspect any warnings.
Download daily weather CSV	Download	Depends on the active data and prior selections.	Downloads the current daily weather csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Run phenology simulation	Action button	Depends on the active data and prior selections.	Runs the action named “Run phenology simulation” using the current selections. Terms: Phenology: The timing of crop development stages such as emergence, flowering and maturity.	Review inputs, then run the action and inspect any warnings.
Download complete phenology package	Download	Depends on the active data and prior selections.	Downloads the current complete phenology package for archiving, analysis or reporting. Terms: Phenology: The timing of crop development stages such as emergence, flowering and maturity.	Save it with the project, country and analysis date in the filename.
Build historical season comparison	Action button	Depends on the active data and prior selections.	Runs the action named “Build historical season comparison” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download historical season summary	Download	Depends on the active data and prior selections.	Downloads the current historical season summary for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download target percentiles	Download	Depends on the active data and prior selections.	Downloads the current target percentiles for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Clear NASA POWER daily cache	Action button	Depends on the active data and prior selections.	Performs a destructive or clearing action. Review confirmation controls and backups before using it. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Use only after verifying the active country, project, field or trial.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

Agricultural note

Crop parameters describe a screening or modelling assumption. Replace defaults with locally validated cultivar, soil and management values when available, and document all changes.

8.2 Seasonal suitability

Path: Crop Decisions > Seasonal suitability

Combine a literature-derived crop climate envelope with FAO crop-water coefficients for a selected local growth window.

Worked example

Example: evaluate maize suitability for a May-September window using temperature, precipitation and water-balance limits.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Absolute minimum (°C), Optimal minimum (°C), Optimal maximum (°C), Absolute maximum (°C).
3. Run “Generate suitability map”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Absolute minimum (°C)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Optimal minimum (°C)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Optimal maximum (°C)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Absolute maximum (°C)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Absolute minimum (mm/year or crop-cycle screen)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Optimal minimum (mm)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Optimal maximum (mm)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value

Field/control	Type	Options/default	Meaning	Example
Absolute maximum (mm)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Crop coefficient Kc	Number	Range: 0.01 to 2.5 Default: generated from the current data or context	Enter or adjust the numeric setting “Crop coefficient Kc”. Use documented units and a scientifically justified value. Terms: Kc: Crop coefficient: a stage-dependent multiplier that converts ET ₀ into crop evapotranspiration.	Start with the default, then test sensitivity where appropriate.
Use FAO Ky yield-response factor	Checkbox	Default: generated from the current data or context	Enables or disables “Use FAO Ky yield-response factor”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
Yield-response factor Ky	Number	Range: 0.0 to 3.0 Default: generated from the current data or context	Enter or adjust the numeric setting “Yield-response factor Ky”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Effective precipitation fraction	Slider	Range: 0.0 to 1.0 Default: generated from the current data or context	Controls or selects rainfall-related inputs or outputs, usually expressed in millimetres.	500 mm seasonal rainfall
Legacy water-deficit absolute lower limit (mm)	Number	Default: generated from the current data or context	Enter or adjust the numeric setting “Legacy water-deficit absolute lower limit (mm)”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Legacy water-deficit optimal threshold (mm)	Number	Default: generated from the current data or context	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Crop	Drop-down	Options: generated from the current data or context	Select the crop whose validated parameters or custom profile should be used.	Maize
Growth stage / analysis window	Drop-down	Options: generated from the current data or context	Selects a crop-development phase such as establishment, flowering or maturity.	Flowering
Year	Drop-down	Options: available years	Selects a calendar or season year included in the analysis.	2025
Locally appropriate months	Multi-select	Options: generated from the current data or context Default: generated from the current data or context Built-in help: The built-in month sequence is illustrative, not a validated national planting calendar.	Selects the months included in a crop window or climate summary.	May-September
Component aggregation	Single choice	Options: Limiting factor,	Defines how multiple	Limiting factor

Field/control	Type	Options/default	Meaning	Example
		Mean of components	components are combined, for example by the limiting factor or their mean.	
Location for historical detail	Drop-down	Options: generated from the current data or context	Select one option for "Location for historical detail". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download results CSV	Download	Depends on the active data and prior selections.	Downloads the current results csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Generate suitability map	Action button	Depends on the active data and prior selections.	Runs the action named "Generate suitability map" using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Suitability scores are climate-envelope screening outputs. They do not include all genotype, soil, disease, nutrient or management constraints.

Agricultural note

Crop parameters describe a screening or modelling assumption. Replace defaults with locally validated cultivar, soil and management values when available, and document all changes.

8.3 Suitability stability

Path: Crop Decisions > Suitability stability

Evaluate the consistency of validated climate-envelope and FAO crop-water suitability across years.

Worked example

Example: calculate how often a location remains highly suitable across 30 historical seasons instead of relying on one average year.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.

2. Configure the main inputs: Absolute minimum (°C), Optimal minimum (°C), Optimal maximum (°C), Absolute maximum (°C).
3. Run “Generate stability map”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Absolute minimum (°C)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Optimal minimum (°C)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Optimal maximum (°C)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Absolute maximum (°C)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Absolute minimum (mm/year or crop-cycle screen)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Optimal minimum (mm)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Optimal maximum (mm)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Absolute maximum (mm)	Number	Default: generated from the current data or context	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Crop coefficient Kc	Number	Range: 0.01 to 2.5 Default: generated from the current data or context	Enter or adjust the numeric setting “Crop coefficient Kc”. Use documented units and a scientifically justified value. Terms: Kc: Crop coefficient: a stage-dependent multiplier that converts ET ₀ into crop evapotranspiration.	Start with the default, then test sensitivity where appropriate.
Use FAO Ky yield-response factor	Checkbox	Default: generated from the current data or context	Enables or disables “Use FAO Ky yield-response factor”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
Yield-response factor Ky	Number	Range: 0.0 to 3.0 Default: generated from the current data or context	Enter or adjust the numeric setting “Yield-response factor Ky”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Effective precipitation fraction	Slider	Range: 0.0 to 1.0 Default: generated from the current data or context	Controls or selects rainfall-related inputs or outputs, usually expressed in millimetres.	500 mm seasonal rainfall

Field/control	Type	Options/default	Meaning	Example
Legacy water-deficit absolute lower limit (mm)	Number	Default: generated from the current data or context	Enter or adjust the numeric setting “Legacy water-deficit absolute lower limit (mm)”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Legacy water-deficit optimal threshold (mm)	Number	Default: generated from the current data or context	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Crop	Drop-down	Options: generated from the current data or context	Select the crop whose validated parameters or custom profile should be used.	Maize
Growth stage / window	Drop-down	Options: generated from the current data or context	Selects a crop-development phase such as establishment, flowering or maturity.	Flowering
Locally appropriate months	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Selects the months included in a crop window or climate summary.	May-September
Analysis period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Yearly component aggregation	Single choice	Options: Limiting factor, Mean of components	Defines how multiple components are combined, for example by the limiting factor or their mean.	Limiting factor
Inspect location	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download stability summary	Download	Depends on the active data and prior selections.	Downloads the current stability summary for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Generate stability map	Action button	Depends on the active data and prior selections.	Runs the action named “Generate stability map” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

Agricultural note

Crop parameters describe a screening or modelling assumption. Replace defaults with locally validated cultivar, soil and management values when available, and document all changes.

8.4 Crop planning laboratory

Path: Crop Decisions > Crop planning laboratory

Build crop profiles, optimise planting windows, estimate irrigation demand, diagnose stage vulnerability, compare crop portfolios, and identify substitution options.

Worked example

Example: compare planting windows, irrigation demand and stage vulnerability for maize, wheat and sorghum at one location.

Tabs and views

Tab/view	Purpose
Custom crop builder	Opens the custom crop builder stage or result view.
Planting-date optimiser	Opens the planting-date optimiser stage or result view.
Irrigation demand	Opens the irrigation demand stage or result view.
Stage vulnerability	Opens the stage vulnerability stage or result view.
Multi-crop portfolio	Opens the multi-crop portfolio stage or result view.
Crop substitution	Opens the crop substitution stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Crop name, Stage / seasonal window name, Months, Temperature absolute minimum.
3. Run “Download interpretation as Markdown”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Crop name	Text	Default: New Crop	Enter the requested text for “Crop name”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Stage / seasonal window name	Text	Default: Full growing season	Selects a crop-development phase such	Flowering

Field/control	Type	Options/default	Meaning	Example
			as establishment, flowering or maturity.	
Months	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Selects the months included in a crop window or climate summary.	May-September
Temperature absolute minimum	Number	Default: 8.0	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Temperature optimal minimum	Number	Default: 18.0	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Temperature optimal maximum	Number	Default: 30.0	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Temperature absolute maximum	Number	Default: 40.0	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Water-deficit absolute lower limit	Number	Default: -500.0	Enter or adjust the numeric setting "Water-deficit absolute lower limit". Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Seasonal precipitation absolute minimum	Number	Default: 200.0	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Seasonal precipitation optimal minimum	Number	Default: 400.0	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Seasonal precipitation optimal maximum	Number	Default: 900.0	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Seasonal precipitation absolute maximum	Number	Default: 1400.0	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Water-deficit optimal threshold	Number	Default: 0.0	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Notes / source	Long text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Crop	Drop-down	Options: generated from the current data or context	Select the crop whose validated parameters or custom profile should be used.	Maize
Stage used for the shifted window	Drop-down	Options: generated from the current data or context	Selects a crop-development phase such as establishment, flowering or maturity.	Flowering
Location	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Historical period	Slider	Range: generated from the current data or	Defines the years or dates included in the	1991-2025

Field/control	Type	Options/default	Meaning	Example
		context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	calculation.	
Window length (months)	Slider	Range: 1 to 12 Default: generated from the current data or context	Enter or adjust the numeric setting “Window length (months)”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Suitability aggregation	Single choice	Options: Limiting factor, Mean of components	Defines how multiple components are combined, for example by the limiting factor or their mean.	Limiting factor
Stage / season	Drop-down	Options: generated from the current data or context	Selects a crop-development phase such as establishment, flowering or maturity.	Flowering
Period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Average crop coefficient (Kc)	Slider	Range: 0.2 to 1.6 Default: generated from the current data or context	Enter or adjust the numeric setting “Average crop coefficient (Kc)”. Use documented units and a scientifically justified value. Terms: Kc: Crop coefficient: a stage-dependent multiplier that converts ET ₀ into crop evapotranspiration.	Start with the default, then test sensitivity where appropriate.
Irrigation-system efficiency	Slider	Range: 0.3 to 1.0 Default: 0.75	Configures irrigation timing, amount, efficiency or scenario assumptions.	Trigger at RAW with 75% efficiency
Effective precipitation fraction	Slider	Range: 0.0 to 1.0 Default: 0.8	Controls or selects rainfall-related inputs or outputs, usually expressed in millimetres.	500 mm seasonal rainfall
Aggregation	Single choice	Options: Limiting factor, Mean of components	Defines how multiple components are combined, for example by the limiting factor or their mean.	Limiting factor
Crops in the portfolio	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Crops in the portfolio”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Inspect location	Drop-down	Options: generated from the current data or	Selects the geographic or experimental unit used by	A mapped research field or catalogue city

Field/control	Type	Options/default	Meaning	Example
		context	the analysis.	
Current crop	Drop-down	Options: generated from the current data or context	Select one option for “Current crop”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download all crop profiles JSON	Download	Depends on the active data and prior selections.	Downloads the current all crop profiles json for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

8.5 Current-season monitor

Path: Crop Decisions > Current-season monitor

Combine project phenology, near-real-time NASA weather, root-zone status and the latest processed Sentinel-2 observation in one transparent season dashboard.

Worked example

Example: combine a 1 May planting date, recent NASA weather, the latest root-zone simulation and Sentinel-2 NDVI to see whether the crop is developing and experiencing stress as expected.

Tabs and views

Tab/view	Purpose
Season overview	Opens the season overview stage or result view.
Weather trajectory	Opens the weather trajectory stage or result view.
Root-zone trajectory	Opens the root-zone trajectory stage or result view.
Satellite status	Opens the satellite status stage or result view.
Save & methods	Opens the save & methods stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: GDD base temperature (°C), GDD upper cap (°C), Heat alert threshold (°C), Ignore NASA cache.
3. Run “Download interpretation as Markdown”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
GDD base temperature (°C)	Number	Range: -5.0 to 30.0 Default: generated from the current data or context	Controls or selects a temperature threshold, range or variable in degrees Celsius. Terms: GDD: Growing degree days: accumulated thermal time above a base temperature, usually capped at an upper temperature.	30 °C
GDD upper cap (°C)	Number	Range: 10.0 to 60.0 Default: generated from the current data or context	Enter or adjust the numeric setting “GDD upper cap (°C)”. Use documented units and a scientifically justified value. Terms: GDD: Growing degree days: accumulated thermal time above a base temperature, usually capped at an upper temperature.	Start with the default, then test sensitivity where appropriate.
Heat alert threshold (°C)	Number	Range: 20.0 to 55.0 Default: 35.0	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Ignore NASA cache	Checkbox	Default: False	Enables or disables “Ignore NASA cache”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Refresh current-season weather and root-zone status	Action button	Depends on the active data and prior selections.	Runs the action named “Refresh current-season weather and root-zone status” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download live snapshot JSON	Download	Depends on the active data and prior selections.	Downloads the current live snapshot json for archiving, analysis or	Save it with the project, country and analysis date in the filename.

Action	Type	Conditions/default	What it does	Good practice
			reporting.	
Save live snapshot to active project	Action button	Depends on the active data and prior selections.	Runs the action named “Save live snapshot to active project” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

8.6 Soil-water balance

Path: Crop Decisions > Soil-water balance

Model daily crop water use, root-zone depletion, agricultural water stress, runoff, deep percolation and irrigation scheduling using the validated crop library and NASA POWER daily weather.

Worked example

Example: simulate maize in a loam soil with 65% irrigation efficiency and irrigate when depletion reaches RAW; compare stress days with rainfed management.

Tabs and views

Tab/view	Purpose
Single season	Run one crop-season root-zone simulation.
Historical risk	Repeat the water-balance model across historical seasons.
Scenario comparison	Compare baseline soil-water management with an intervention.
Soil profiles & methods	Opens the soil profiles & methods stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Location source, Location, Latitude, Longitude.
3. Run “Run root-zone simulation”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Location source	Single choice	Options: App city, Custom coordinates	Choose a location from the country catalogue or enter exact custom coordinates.	App city
Location	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city

Field/control	Type	Options/default	Meaning	Example
Latitude	Number	Range: -90.0 to 90.0 Default: generated from the current data or context	North-south coordinate in decimal degrees. Positive values are north of the equator.	19.46
Longitude	Number	Range: -180.0 to 180.0 Default: generated from the current data or context	East-west coordinate in decimal degrees. Mexico and Cyprus longitudes differ in sign; verify the map before downloading data.	-98.90
Crop	Drop-down	Options: available crops	Select the crop whose validated parameters or custom profile should be used.	Maize
Water-parameter profile	Drop-down	Options: available parameter profiles	Select the crop parameter record containing crop coefficients, depletion fraction and root-depth assumptions. Terms: Kc: Crop coefficient: a stage-dependent multiplier that converts ET ₀ into crop evapotranspiration. Root-zone depletion: The amount of plant-available water removed from the effective rooting depth relative to field capacity.	Validated maize profile
Stage-duration value	Drop-down	Options: Midpoint, Minimum, Maximum	Choose the minimum, midpoint or maximum duration from a published range.	Midpoint
User-defined simulation length (days)	Number	Range: 1 to 730 Default: generated from the current data or context	Enter or adjust the numeric setting “User-defined simulation length (days)”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Constant Kc	Number	Range: 0.05 to 2.0 Default: generated from the current data or context	Enter or adjust the numeric setting “Constant Kc”. Use documented units and a scientifically justified value. Terms: Kc: Crop coefficient: a stage-dependent multiplier that converts ET ₀ into crop evapotranspiration.	Start with the default, then test sensitivity where appropriate.
Planting or season start date	Date	Default: generated from the current data or context	Sets day 1 for weather, phenology and crop-water calculations. Terms: Phenology: The timing of crop development stages such as emergence, flowering and maturity.	1 May 2027
Initial effective root depth (m)	Number	Range: 0.05 to 3.0 Default: generated from the current data or context	Effective rooting depth at the beginning of the simulation.	0.20 m
Maximum effective root depth (m)	Number	Range: 0.1 to 4.0 Default: generated from the current data or context	Largest root-zone depth available to the crop in the model.	1.20 m

Field/control	Type	Options/default	Meaning	Example
Fallback depletion fraction p	Number	Range: 0.1 to 0.8 Default: generated from the current data or context	Fraction of TAW that is readily available before stress when a stage-specific value is unavailable. Terms: TAW: Total available water: root-zone water held between field capacity and permanent wilting point. Root-zone depletion: The amount of plant-available water removed from the effective rooting depth relative to field capacity.	0.50
Dynamic root growth	Checkbox	Default: True Built-in help: Root depth increases from the initial value to the selected maximum during approximately the first half of the crop season.	When enabled, root depth expands from the initial to maximum depth during early crop development.	Enabled
Soil-water input	Single choice	Options: FAO screening preset, Custom field capacity and wilting point	Choose a screening soil-texture preset or enter measured field capacity and wilting point. Terms: Field capacity: The water content remaining after excess gravitational water has drained from the soil. Permanent wilting point: The soil water content below which plants generally cannot recover turgor.	Custom field capacity and wilting point
Soil texture preset	Drop-down	Options: generated from the current data or context	Select one option for "Soil texture preset". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Volumetric water content at field capacity ($\text{m}^3 \text{m}^{-3}$)	Number	Range: 0.01 to 0.7 Default: 0.3	Enter soil water content at field capacity as a volume fraction. Terms: Field capacity: The water content remaining after excess gravitational water has drained from the soil.	0.30
Volumetric water content at permanent wilting point ($\text{m}^3 \text{m}^{-3}$)	Number	Range: 0.001 to 0.6 Default: 0.15	Enter soil water content at permanent wilting point as a volume fraction. Terms: Permanent wilting point: The soil water content below which plants generally cannot recover turgor.	0.15

Field/control	Type	Options/default	Meaning	Example
Initial depletion (% of TAW)	Slider	Range: 0 to 100 Default: 20	Sets how dry the root zone is at the simulation start. Terms: TAW: Total available water: root-zone water held between field capacity and permanent wilting point. Root-zone depletion: The amount of plant-available water removed from the effective rooting depth relative to field capacity.	25%
Runoff method	Drop-down	Options: None, Fixed fraction, NRCS curve number	Choose how rainfall runoff is estimated: none, a fixed fraction or the NRCS curve-number method.	NRCS curve number
Runoff fraction of rainfall	Slider	Range: 0.0 to 0.9 Default: 0.0	Controls or selects rainfall-related inputs or outputs, usually expressed in millimetres.	500 mm seasonal rainfall
NRCS curve number	Number	Range: 30 to 100 Default: 75	Enter or adjust the numeric setting “NRCS curve number”. Use documented units and a scientifically justified value. Terms: Curve number: An NRCS runoff parameter representing rainfall-runoff potential; higher values generally imply more runoff.	Start with the default, then test sensitivity where appropriate.
Capillary rise (mm day ⁻¹)	Number	Range: 0.0 to 20.0 Default: 0.0 Built-in help: Keep at zero unless shallow-groundwater contribution is supported by local evidence.	Enter or adjust the numeric setting “Capillary rise (mm day ⁻¹)”. Use documented units and a scientifically justified value. Terms: Capillary rise: Upward movement of groundwater into the root zone; it should be used only when supported by local evidence.	Start with the default, then test sensitivity where appropriate.
Adjust p for daily ETc	Checkbox	Default: True Built-in help: Applies the FAO-56 approximation $p = p_{table} + 0.04(5 - ET_c)$, constrained to 0.1–0.8.	Enables or disables “Adjust p for daily ETc”. Leave it at the default unless the alternative is intentional. Terms: ETc: Crop evapotranspiration: estimated crop water use, commonly calculated as crop coefficient Kc multiplied by ET0.	Toggle only when the displayed workflow requires it.
Irrigation strategy	Drop-down	Options: generated from the current data or context	Configures irrigation timing, amount, efficiency or scenario assumptions.	Trigger at RAW with 75% efficiency
Application efficiency	Slider	Range: 0.2 to 1.0 Default: 0.75	Fraction of gross irrigation that reaches and is stored in the root zone.	0.75
Trigger (× RAW)	Number	Range: 0.1 to 2.0 Default:	Controls when irrigation	1.0

Field/control	Type	Options/default	Meaning	Example
		1.0	starts relative to readily available water. A value of 1 triggers near the RAW threshold. Terms: RAW: Readily available water: the fraction of TAW that can be depleted before crop water stress is expected.	
Fraction of current depletion refilled	Slider	Range: 0.0 to 1.0 Default: generated from the current data or context	Fraction of the current root-zone deficit replaced by each triggered irrigation. Terms: Root-zone depletion: The amount of plant-available water removed from the effective rooting depth relative to field capacity.	1.0
Maximum gross application per event (mm)	Number	Range: 0.0 to 300.0 Default: 60.0	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Interval (days)	Number	Range: 1 to 60 Default: 7	Enter or adjust the numeric setting "Interval (days)". Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Gross application per event (mm)	Number	Range: 0.0 to 300.0 Default: 25.0	Enter or adjust the numeric setting "Gross application per event (mm)". Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Upload irrigation schedule CSV	File upload	Built-in help: Required columns: Date, Gross irrigation (mm) Accepted: csv	Uploads the file required for upload irrigation schedule csv. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Ignore NASA POWER cache and download again	Checkbox	Default: False	Enables or disables "Ignore NASA POWER cache and download again". Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
First season year	Number	Range: 1981 to generated from the current data or context Default: generated from the current data or context	Selects a calendar or season year included in the analysis.	2025
Last season year	Number	Range: 1981 to generated from the current data or context Default: generated from the current data or context	Selects a calendar or season year included in the analysis.	2025
Target season year	Number	Range: 1981 to generated from the current data or context Default: generated from the current data or context	The season to compare against historical years or a baseline.	2025
Minimum season completeness	Slider	Range: 50 to 100 Default: 90	Sets a lower or upper screening, processing or	Use a locally justified value

Field/control	Type	Options/default	Meaning	Example
			validation limit.	
Redownload historical daily weather	Checkbox	Default: False	Enables or disables “Redownload historical daily weather”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
Historical chart metric	Drop-down	Options: generated from the current data or context	Selects the result variable used in a chart, ranking or comparison.	RMSE
Available-water-capacity multiplier	Slider	Range: 0.5 to 2.0 Default: 1.2	Enter or adjust the numeric setting “Available-water-capacity multiplier”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Maximum-root-depth multiplier	Slider	Range: 0.5 to 1.5 Default: 1.0	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Intervention irrigation	Drop-down	Options: Rainfed, Irrigate at RAW, Deficit irrigation, Fixed interval	Configures irrigation timing, amount, efficiency or scenario assumptions.	Trigger at RAW with 75% efficiency
Intervention efficiency	Slider	Range: 0.2 to 1.0 Default: 0.85	Fraction of an applied resource that reaches its intended target.	0.75
Comparison metrics	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Selects the result variable used in a chart, ranking or comparison.	RMSE

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download daily balance CSV	Download	Depends on the active data and prior selections.	Downloads the current daily balance csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download stage summary CSV	Download	Depends on the active data and prior selections.	Downloads the current stage summary csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download complete package	Download	Depends on the active data and prior selections.	Downloads the current complete package for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Run root-zone simulation	Action button	Depends on the active data and prior selections.	Runs the action named “Run root-zone simulation” using the current selections.	Review inputs, then run the action and inspect any warnings.
Run historical soil-water analysis	Action button	Depends on the active data and prior selections.	Runs the action named “Run historical soil-water analysis” using the	Review inputs, then run the action and inspect any warnings.

Action	Type	Conditions/default	What it does	Good practice
			current selections.	
Download historical seasons	Download	Depends on the active data and prior selections.	Downloads the current historical seasons for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download target percentiles	Download	Depends on the active data and prior selections.	Downloads the current target percentiles for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Compare baseline and intervention	Action button	Depends on the active data and prior selections.	Runs the action named “Compare baseline and intervention” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download scenario comparison	Download	Depends on the active data and prior selections.	Downloads the current scenario comparison for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download soil-profile template	Download	Depends on the active data and prior selections.	Downloads the current soil-profile template for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download irrigation-schedule template	Download	Depends on the active data and prior selections.	Downloads the current irrigation-schedule template for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

Depletion, Ks and irrigation requirements depend strongly on soil, root depth, crop coefficient and irrigation assumptions. Review them before acting.

Agricultural note

Crop parameters describe a screening or modelling assumption. Replace defaults with locally validated cultivar, soil and management values when available, and document all changes.

8.7 Yield calculator

Path: Crop Decisions > Yield calculator

Calculate a transparent relative climate-yield index and optionally scale it by a user-supplied potential yield.

Worked example

Example: convert a relative climate-yield index of 0.78 into an indicative 7.8 t/ha when potential yield is set to 10 t/ha.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Crop, Year, Period, Output.
3. Run “Download interpretation as Markdown”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Crop	Drop-down	Options: generated from the current data or context	Select the crop whose validated parameters or custom profile should be used.	Maize
Year	Drop-down	Options: available years	Selects a calendar or season year included in the analysis.	2025
Period	Drop-down	Options: generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Output	Single choice	Options: Relative climate-yield index, Scale by user-supplied potential yield	Select one option for "Output". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Potential yield for the chosen cultivar/system (t/ha)	Number	Range: 0.01 to context dependent Default: 5.0	Identifies the genetic material being studied. Use the official breeding or cultivar code.	CML-XXX

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download yield-response results	Download	Depends on the active data and prior selections.	Downloads the current yield-response results for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

The output is an indicative climate-limited yield screen, not a calibrated yield forecast unless validated with local observations.

Agricultural note

Crop parameters describe a screening or modelling assumption. Replace defaults with locally validated cultivar, soil and management values when available, and document all changes.

8.8 Water productivity & economics

Path: Crop Decisions > Water productivity & economics

Translate yield, evapotranspiration and irrigation scenarios into water-productivity, cost, gross-margin and financial-risk indicators.

Worked example

Example: compare two irrigation scenarios for yield per cubic metre, gross margin and probability of financial loss.

Tabs and views

Tab/view	Purpose
Add scenario	Opens the add scenario stage or result view.
Scenario comparison	Compare baseline soil-water management with an intervention.
Financial uncertainty	Opens the financial uncertainty stage or result view.
Export & project	Opens the export & project stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Scenario name, Baseline scenario, Remove scenarios, Scenario for Monte Carlo uncertainty.
3. Run “Run economic uncertainty analysis”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Scenario name	Text	Default: {season.get('crop', 'Crop')} baseline	Selects or defines an alternative assumption set for comparison with the baseline.	Add 25 mm irrigation
Baseline scenario	Drop-down	Options: generated from the current data or context	Defines the reference period or scenario against which change is measured.	1985-2014
Remove scenarios	Multi-select	Options: generated from the current data or context	Selects or defines an alternative assumption set for comparison with the baseline.	Add 25 mm irrigation
Scenario for Monte Carlo uncertainty	Drop-down	Options: generated from the current data or context	Selects or defines an alternative assumption set for comparison with the baseline.	Add 25 mm irrigation
Iterations	Selection slider	Range: 500, 1000, 2500, 5000, 10000 to context dependent Default: 5000	Enter or adjust the numeric setting “Iterations”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Remove selected scenarios	Action button	Depends on the active data and prior selections.	Runs the action named "Remove selected scenarios" using the current selections.	Review inputs, then run the action and inspect any warnings.
Run economic uncertainty analysis	Action button	Depends on the active data and prior selections.	Runs the action named "Run economic uncertainty analysis" using the current selections.	Review inputs, then run the action and inspect any warnings.
Download economics and water-productivity package	Download	Depends on the active data and prior selections.	Downloads the current economics and water-productivity package for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Save economic analysis to active project	Action button	Depends on the active data and prior selections.	Runs the action named "Save economic analysis to active project" using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Economic results depend on the entered prices, costs, irrigation quantities and yield assumptions; run sensitivity analyses before decisions.

8.9 AquaCrop simulations

Path: Crop Decisions > AquaCrop simulations

Prepare project-specific daily weather and run AquaCrop-OSPy crop-water simulations with transparent soil, crop and irrigation configuration.

Worked example

Example: prepare daily weather for a maize project, select a soil and irrigation strategy, run AquaCrop-OSPy and compare simulated yield and water use.

Tabs and views

Tab/view	Purpose
Weather preparation	Opens the weather preparation stage or result view.
Simulation settings	Opens the simulation settings stage or result view.
Results	Opens the results stage or result view.
Methods & provenance	Opens the methods & provenance stage or result view.
Final statistics	Opens the final statistics stage or result view.
Crop growth	Opens the crop growth stage or result view.
Water flux	Opens the water flux stage or result view.
Water storage	Opens the water storage stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Daily weather source, Ignore NASA POWER cache, AquaCrop crop name, AquaCrop soil type.
3. Run “Run AquaCrop-OSPy simulation”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Daily weather source	Drop-down	Options: generated from the current data or context	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Ignore NASA POWER cache	Checkbox	Default: False	Enables or disables “Ignore NASA POWER cache”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
AquaCrop crop name	Text	Default: generated from the current data or context	Enter the requested text for “AquaCrop crop name”. Use consistent identifiers and document methods or sources where relevant. Terms: AquaCrop: FAO’s crop-water productivity model, represented here through AquaCrop-OSPy when installed.	Use a clear, stable name that will remain meaningful in exports.
AquaCrop soil type	Drop-down	Options: generated from the current data or context	Select one option for “AquaCrop soil type”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects. AquaCrop: FAO’s crop-water productivity model, represented here through AquaCrop-OSPy when installed.	Choose the option matching the research question.
Initial water content	Drop-down	Options: FC, WP, SAT	Select one option for “Initial water content”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			reduce some background and atmospheric effects.	
Irrigation management	Drop-down	Options: generated from the current data or context	Configures irrigation timing, amount, efficiency or scenario assumptions.	Trigger at RAW with 75% efficiency
Application efficiency (%)	Number	Range: 1.0 to 100.0 Default: 75.0	Fraction of an applied resource that reaches its intended target.	0.75
Maximum daily irrigation (mm)	Number	Range: 0.0 to 200.0 Default: 25.0	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Stage soil-moisture targets (% TAW)	Multi-select	Options: 40, 50, 60, 70, 80, 90, 100 Default: 70	Selects a crop-development phase such as establishment, flowering or maturity. Terms: TAW: Total available water: root-zone water held between field capacity and permanent wilting point.	Flowering
Irrigation interval (days)	Number	Range: 1 to 60 Default: 7	Configures irrigation timing, amount, efficiency or scenario assumptions.	Trigger at RAW with 75% efficiency
Net-irrigation target (% TAW)	Number	Range: 1.0 to 100.0 Default: 70.0	Selects the response variable or event the analysis predicts or evaluates. Terms: TAW: Total available water: root-zone water held between field capacity and permanent wilting point.	Seed-set percentage
Constant daily depth (mm)	Number	Range: 0.0 to 100.0 Default: 10.0	Enter or adjust the numeric setting “Constant daily depth (mm)”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Irrigation schedule CSV with Date and Depth	File upload	Accepted: csv	Uploads the file required for irrigation schedule csv with date and depth. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Simulation start	Date	Default: generated from the current data or context	Sets the date or time for “Simulation start”. Confirm that it matches the crop season and local field records.	Use the recorded field date rather than an approximate date when available.
Simulation end	Date	Default: generated from the current data or context	Sets the date or time for “Simulation end”. Confirm that it matches the crop season and local field records.	Use the recorded field date rather than an approximate date when available.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Prepare AquaCrop weather	Action button	Depends on the active data and prior selections.	Runs the action named “Prepare AquaCrop weather” using the current selections. Terms: AquaCrop: FAO’s crop-water productivity model, represented here through AquaCrop-OSPy when installed.	Review inputs, then run the action and inspect any warnings.
Download weather and configuration package	Download	Depends on the active data and prior selections.	Downloads the current weather and configuration package for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Run AquaCrop-OSPy simulation	Action button	Depends on the active data and prior selections.	Runs the action named “Run AquaCrop-OSPy simulation” using the current selections. Terms: AquaCrop: FAO’s crop-water productivity model, represented here through AquaCrop-OSPy when installed.	Review inputs, then run the action and inspect any warnings.
Download complete AquaCrop run package	Download	Depends on the active data and prior selections.	Downloads the current complete aquacrop run package for archiving, analysis or reporting. Terms: AquaCrop: FAO’s crop-water productivity model, represented here through AquaCrop-OSPy when installed.	Save it with the project, country and analysis date in the filename.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

8.10 DSSAT & APSIM interoperability

Path: Crop Decisions > DSSAT & APSIM interoperability

Prepare traceable weather packages, execute locally installed model templates, and import model outputs without silently rewriting fragile experiment definitions.

Worked example

Example: export a DSSAT weather file, run a locally prepared experiment template and import the model output for comparison with field observations.

Tabs and views

Tab/view	Purpose
Weather	Opens the weather stage or result view.
Prepare packages	Opens the prepare packages stage or result view.
Run DSSAT	Opens the run dssat stage or result view.
Run APSIM	Opens the run apsim stage or result view.
Outputs & project	Opens the outputs & project stage or result view.
Methods	Opens the methods stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Daily weather source, Upload daily CSV containing date, minimum temperature, maximum temperature, rainfall and solar radiation, Four-character DSSAT station code, Optional APSIMX template.
3. Run “Generate DSSAT and APSIM input packages”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Daily weather source	Single choice	Options: Use current app daily weather, Upload daily-weather CSV	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Upload daily CSV containing date, minimum temperature, maximum temperature, rainfall and solar radiation	File upload	Accepted: csv	Uploads the file required for upload daily csv containing date, minimum temperature, maximum temperature, rainfall and solar radiation. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Four-character DSSAT station code	Text	Default: MX01	Enter the requested text for “Four-character DSSAT station code”. Use consistent identifiers and document methods or sources where relevant. Terms: DSSAT: Decision Support System for Agrotechnology Transfer, an external process-based crop-modelling system.	Use a clear, stable name that will remain meaningful in exports.
Optional APSIMX template	File upload	Accepted: apsimx	Uploads the file required for optional apsimx template. Check the displayed template or required column names before import. Terms: APSIM: Agricultural	Use the matching template supplied by AgroLattice.

Field/control	Type	Options/default	Meaning	Example
			Production Systems sIMulator, an external process-based modelling framework.	
DSSAT executable path	Text	Default: generated from the current data or context	Enter the requested text for “DSSAT executable path”. Use consistent identifiers and document methods or sources where relevant. Terms: DSSAT: Decision Support System for Agrotechnology Transfer, an external process-based crop-modelling system.	Use a clear, stable name that will remain meaningful in exports.
Upload prepared DSSAT run package (.zip)	File upload	Accepted: zip	Uploads the file required for upload prepared dssat run package (.zip). Check the displayed template or required column names before import. Terms: DSSAT: Decision Support System for Agrotechnology Transfer, an external process-based crop-modelling system.	Use the matching template supplied by AgroLattice.
DSSAT timeout (seconds)	Number	Range: 30 to 86400 Default: 1800	Enter or adjust the numeric setting “DSSAT timeout (seconds)”. Use documented units and a scientifically justified value. Terms: DSSAT: Decision Support System for Agrotechnology Transfer, an external process-based crop-modelling system.	Start with the default, then test sensitivity where appropriate.
APSIM Models.exe path	Text	Default: generated from the current data or context	Enter the requested text for “APSIM Models.exe path”. Use consistent identifiers and document methods or sources where relevant. Terms: APSIM: Agricultural Production Systems sIMulator, an external process-based modelling framework.	Use a clear, stable name that will remain meaningful in exports.
Upload APSIM run package (.zip)	File upload	Accepted: zip	Uploads the file required for upload apsim run package (.zip). Check the displayed template or required column names before import. Terms: APSIM: Agricultural Production Systems sIMulator, an external process-based modelling framework.	Use the matching template supplied by AgroLattice.
Run through APSIM command language	Checkbox	Default: True	Enables or disables “Run through APSIM command language”. Leave it at the default unless the	Toggle only when the displayed workflow requires it.

Field/control	Type	Options/default	Meaning	Example
			alternative is intentional. Terms: APSIM: Agricultural Production Systems sIMulator, an external process-based modelling framework.	
Model output	Single choice	Options: DSSAT, APSIM	Select one option for “Model output”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Parsed table	Drop-down	Options: generated from the current data or context	Select one option for “Parsed table”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Generate DSSAT and APSIM input packages	Action button	Depends on the active data and prior selections.	Runs the action named “Generate DSSAT and APSIM input packages” using the current selections. Terms: DSSAT: Decision Support System for Agrotechnology Transfer, an external process-based crop-modelling system. APSIM: Agricultural Production Systems sIMulator, an external process-based modelling framework.	Review inputs, then run the action and inspect any warnings.
Prepare DSSAT working directory	Action button	Depends on the active data and prior selections.	Runs the action named “Prepare DSSAT working directory” using the current selections. Terms: DSSAT: Decision Support System for Agrotechnology Transfer, an external process-based crop-modelling system.	Review inputs, then run the action and inspect any warnings.

Action	Type	Conditions/default	What it does	Good practice
Run DSSAT	Action button	Depends on the active data and prior selections.	Runs the action named “Run DSSAT” using the current selections. Terms: DSSAT: Decision Support System for Agrotechnology Transfer, an external process-based crop-modelling system.	Review inputs, then run the action and inspect any warnings.
Prepare APSIM working directory	Action button	Depends on the active data and prior selections.	Runs the action named “Prepare APSIM working directory” using the current selections. Terms: APSIM: Agricultural Production Systems sIMulator, an external process-based modelling framework.	Review inputs, then run the action and inspect any warnings.
Run APSIM	Action button	Depends on the active data and prior selections.	Runs the action named “Run APSIM” using the current selections. Terms: APSIM: Agricultural Production Systems sIMulator, an external process-based modelling framework.	Review inputs, then run the action and inspect any warnings.
Download parsed run package	Download	Depends on the active data and prior selections.	Downloads the current parsed run package for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Save external-model run to active project	Action button	Depends on the active data and prior selections.	Runs the action named “Save external-model run to active project” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

9. Experiments

This workspace supports reproducible field research, especially maize parental-line flowering synchrony, mapped plots, repeated observations and predictive modelling.

Workspace selector

Open Experiments from the sidebar. Use the Workspace view selector and, where present, the Analysis selector to move between related functions without leaving the workspace.

9.1 Maize flowering trials & field data

Path: Experiments > Maize flowering trials & field data

Creates and manages a mapped maize flowering-synchrony trial, randomises sowing-offset treatments, records flowering and growth measurements, stores harvest outcomes and links exact plot geometries to satellite data.

Worked example

Example: create seven male sowing offsets (-6 to +6 days), three blocks and one replicate per block, draw 21 plots, randomise treatments and record daily pollen shed, silking and plant height.

Tabs and views

Tab/view	Purpose
Trial library	Select, inspect or delete saved trials.
Trial setup	Enter parent lines, design, sowing offsets, GDD settings and protocol information.
Plot map & randomisation	Draw plots inside the experiment field and assign treatments reproducibly.
Daily flowering data	Import or enter repeated pollen, silking, height and stress observations.
Flowering dates	Store protocol-defined initiation and flowering dates for male and female parents.
Harvest & seed quality	Record ears, kernels, seed set, yield, kernel rows, pure seed, germination and genetic purity.
Satellite linkage	Send selected plot geometry to satellite monitoring and attach matching results.
Export & methods	Opens the export & methods stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Active pollination trial, Experiment-field source, Upload experiment-field Polygon or MultiPolygon, Trial.
3. Run “Delete selected trial”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Active pollination trial	Drop-down	Options: generated from available data	Select the trial whose plots, observations and outcomes should be used.	2027 maize flowering synchrony trial
Experiment-field source	Single choice	Options: generated from the current data or context	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Upload experiment-field Polygon or MultiPolygon	File upload	Accepted: json, geojson	Uploads the file required for upload experiment-field polygon or multipolygon. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Trial	Drop-down	Options: generated from the current data or context	Select one option for "Trial". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Primary outcome	Drop-down	Options: Seed-set percentage, Seed yield (t/ha), Kernels per ear, Kernel rows per ear, Pure seed (%), Germination (%), Genetic purity (%), Flowering overlap score	Choose the main response the experiment is designed to optimise or explain.	Seed-set percentage
Protocol notes	Long text	Depends on the active data and prior selections.	Document design details, sampling rules, measurement definitions, exclusions and deviations from protocol. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Assess 20 male and 20 female plants per plot each morning.
Male sowing offsets relative to female (days)	Text	Default: -6,-4,-2,0,2,4,6 Built-in help: Negative values sow the male before the female; positive values sow it after the female.	Enter comma-separated male planting offsets relative to the female sowing date. Negative values plant the male earlier; positive values plant it later.	-6,-4,-2,0,2,4,6
Randomisation seed	Number	Range: 0 to 2147483647 Default: 2027	A fixed number that makes the random treatment assignment reproducible. Reusing the same seed and design gives the same assignment.	2027
Upload completed daily flowering CSV	File upload	Accepted: csv	Uploads the file required for upload completed daily flowering csv. Check the displayed template or required column names	Use the matching template supplied by AgroLattice.

Field/control	Type	Options/default	Meaning	Example
			before import.	
Notes	Text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Upload completed flowering-date CSV	File upload	Accepted: csv	Uploads the file required for upload completed flowering-date csv. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Plot	Drop-down	Options: generated from the current data or context	Select one option for "Plot". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Definition / notes	Text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Upload completed harvest CSV	File upload	Accepted: csv	Uploads the file required for upload completed harvest csv. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Harvest / laboratory notes	Text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Selected mapped plot(s)	Multi-select	Options: generated from the current data or context	Select one or more entries for "Selected mapped plot(s)". Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Delete selected trial	Action button	Depends on the active data and prior selections.	Performs a destructive or clearing action. Review confirmation controls and backups before using it. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Use only after verifying the active country, project, field or trial.
Randomise treatments and save plot map	Action button	Depends on the active data and prior selections.	Runs the action named "Randomise treatments and save plot map" using	Review inputs, then run the action and inspect any warnings.

Action	Type	Conditions/default	What it does	Good practice
			the current selections.	
Download daily flowering sheet	Download	Depends on the active data and prior selections.	Downloads the current daily flowering sheet for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Import or update flowering observations	Action button	Depends on the active data and prior selections.	Runs the action named "Import or update flowering observations" using the current selections.	Review inputs, then run the action and inspect any warnings.
Download flowering-date sheet	Download	Depends on the active data and prior selections.	Downloads the current flowering-date sheet for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Import or update flowering dates	Action button	Depends on the active data and prior selections.	Runs the action named "Import or update flowering dates" using the current selections.	Review inputs, then run the action and inspect any warnings.
Download harvest sheet	Download	Depends on the active data and prior selections.	Downloads the current harvest sheet for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Import or update harvest outcomes	Action button	Depends on the active data and prior selections.	Runs the action named "Import or update harvest outcomes" using the current selections.	Review inputs, then run the action and inspect any warnings.
Send selected plot(s) to Satellite crop monitoring	Action button	Depends on the active data and prior selections.	Runs the action named "Send selected plot(s) to Satellite crop monitoring" using the current selections.	Review inputs, then run the action and inspect any warnings.
Attach current satellite time series to selected plot(s)	Action button	Depends on the active data and prior selections.	Runs the action named "Attach current satellite time series to selected plot(s)" using the current selections.	Review inputs, then run the action and inspect any warnings.
Download complete trial package	Download	Depends on the active data and prior selections.	Downloads the current complete trial package for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

Agricultural note

Male pollen availability and female silk receptivity must overlap. A calendar-day offset may change between seasons, so record both dates and GDD. Detasselling quality, water stress, heat, plant population and parental genetics can influence seed set and purity.

9.2 Maize synchrony analysis & prediction

Path: Experiments > Maize synchrony analysis & prediction

Calculates male pollen-shed and female silking curves, GDD, synchrony gaps and overlap metrics; then fits predictive models, optimises male sowing offset and produces in-season forecasts.

Worked example

Example: model seed-set percentage using sowing offset, flowering overlap, heat exposure and root-zone stress, then estimate the best tested male sowing interval.

Tabs and views

Tab/view	Purpose
Synchrony curves	Visualise male pollen-shed and female silking/receptivity trajectories.
Weather & GDD	Link daily weather and calculate parent-specific thermal time.
Model table	Opens the model table stage or result view.
Predictive models	Fit and cross-validate candidate models.
Offset optimiser	Evaluate candidate male sowing offsets using a fitted model.
In-season forecast	Estimate remaining GDD and likely event dates.
Export & methods	Opens the export & methods stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Active pollination trial, Or upload daily weather CSV, Treatments, Prediction target.
3. Run “Fit and cross-validate models”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Active pollination trial	Drop-down	Options: generated from available data	Select the trial whose plots, observations and outcomes should be used.	2027 maize flowering synchrony trial
Or upload daily weather CSV	File upload	Accepted: csv	Uploads the file required for or upload daily weather csv. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Treatments	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Treatments”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Prediction target	Drop-down	Options: available targets	Select the outcome the machine-learning models should predict.	Seed-set percentage

Field/control	Type	Options/default	Meaning	Example
Predictors	Multi-select	Options: available predictors Default: generated from the current data or context	Select explanatory variables supplied to the model. Avoid predictors measured after the outcome or derived from the target.	Sowing offset, GDD, overlap score and pre-flowering water stress
Independent validation group	Drop-down	Options: available grouping variables	Choose the grouping variable that defines independent validation units.	Year or field
Validation folds	Slider	Range: 2 to 10 Default: 5	Set the number of cross-validation partitions. The number cannot exceed the available independent groups.	5
Model	Drop-down	Options: generated from the current data or context	Select one option for “Model”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Forecast event	Single choice	Options: Male 50% pollen shed, Female 50% silking	Select one option for “Forecast event”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Target GDD	Number	Range: 1.0 to context dependent Default: generated from the current data or context	Enter the thermal-time requirement for the selected flowering event. Terms: GDD: Growing degree days: accumulated thermal time above a base temperature, usually capped at an upper temperature.	650 GDD
Plot	Drop-down	Options: generated from the current data or context	Select one option for “Plot”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Store this weather with the trial	Action button	Depends on the active data and prior selections.	Runs the action named “Store this weather with the trial” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download modelling table	Download	Depends on the active data and prior selections.	Downloads the current modelling table for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Fit and cross-validate models	Action button	Depends on the active data and prior selections.	Runs the action named “Fit and cross-validate models” using the current selections.	Review inputs, then run the action and inspect any warnings.
Optimise sowing offset	Action button	Depends on the active data and prior selections.	Runs the action named “Optimise sowing offset” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download synchrony and prediction package	Download	Depends on the active data and prior selections.	Downloads the current synchrony and prediction package for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

The preferred sowing offset should be selected inside the range tested experimentally and reported with uncertainty and grouped validation.

Agricultural note

Male pollen availability and female silk receptivity must overlap. A calendar-day offset may change between seasons, so record both dates and GDD. Detasselling quality, water stress, heat, plant population and parental genetics can influence seed set and purity.

9.3 Projects

Path: Experiments > Projects

Create reusable field-season records and preserve crop, location, soil, irrigation, satellite and model-run provenance across modules.

Worked example

Example: create a project named “Montecillo maize 2027” with planting date, crop, coordinates, soil profile and irrigation assumptions, then reuse it in phenology, water balance and satellite modules.

Tabs and views

Tab/view	Purpose
Create project	Opens the create project stage or result view.
Project library	Opens the project library stage or result view.
Active project	Opens the active project stage or result view.
Import & methods	Opens the import & methods stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Location source, Location, Latitude, Longitude.
3. Run “Move selected project to trash”.
4. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Location source	Single choice	Options: App city, Custom coordinates	Choose a location from the country catalogue or enter exact custom coordinates.	App city
Location	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Latitude	Number	Range: -90.0 to 90.0 Default: generated from the current data or context	North-south coordinate in decimal degrees. Positive values are north of the equator.	19.46
Longitude	Number	Range: -180.0 to 180.0 Default: generated from the current data or context	East-west coordinate in decimal degrees. Mexico and Cyprus longitudes differ in sign; verify the map before downloading data.	-98.90
Project name	Text	Default: Montecillo maize season	Enter the requested text for “Project name”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Description	Long text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Crop	Drop-down	Options: generated from the current data or context	Select the crop whose validated parameters or custom profile should be used.	Maize
Planting or season start	Date	Default: generated from the current data or context	Sets the date or time for “Planting or season start”. Confirm that it matches the crop season and local field records.	Use the recorded field date rather than an approximate date when available.
Expected season length (days)	Number	Range: 1 to 1000 Default: 140	Enter or adjust the numeric setting “Expected season length (days)”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Attach the field geometry currently defined in Satellite crop monitoring	Checkbox	Default: generated from the current data or context	Enables or disables “Attach the field geometry currently defined in Satellite crop monitoring”. Leave it at the default unless the alternative is	Toggle only when the displayed workflow requires it.

Field/control	Type	Options/default	Meaning	Example
			intentional.	
Project	Drop-down	Options: generated from the current data or context	Select one option for “Project”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Name	Text	Default: generated from the current data or context	Enter the requested text for “Name”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Status	Drop-down	Options: Active, Completed, Paused, Archived	Select one option for “Status”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Planting date	Date	Default: generated from the current data or context	Sets the date or time for “Planting date”. Confirm that it matches the crop season and local field records.	Use the recorded field date rather than an approximate date when available.
Expected harvest	Date	Default: generated from the current data or context	Sets the date or time for “Expected harvest”. Confirm that it matches the crop season and local field records.	Use the recorded field date rather than an approximate date when available.
Cultivar	Text	Default: generated from the current data or context	Identifies the genetic material being studied. Use the official breeding or cultivar code.	CML-XXX
Location label	Text	Default: generated from the current data or context	Enter the requested text for “Location label”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Description	Long text	Default: generated from the current data or context	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Snapshot label	Text	Default: Workspace {datetime.now().strftime('%Y-%m-%d %H:%M')}	Enter the requested text for “Snapshot label”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Import a project JSON file	File upload	Accepted: json	Uploads the file required for import a project json file. Check the displayed template or required	Use the matching template supplied by AgroLattice.

Field/control	Type	Options/default	Meaning	Example
			column names before import.	

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Move selected project to trash	Action button	Depends on the active data and prior selections.	Runs the action named “Move selected project to trash” using the current selections.	Review inputs, then run the action and inspect any warnings.
Save Module A/B/C snapshot to project	Action button	Depends on the active data and prior selections.	Runs the action named “Save Module A/B/C snapshot to project” using the current selections.	Review inputs, then run the action and inspect any warnings.
Import as a new project	Action button	Depends on the active data and prior selections.	Runs the action named “Import as a new project” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

10. Models & Evidence

This workspace evaluates predictive performance, uncertainty, model disagreement, statistical assumptions and scientific robustness.

Workspace selector

Open Models & Evidence from the sidebar. Use the Workspace view selector and, where present, the Analysis selector to move between related functions without leaving the workspace.

10.1 Validation Centre

Path: Models & Evidence > Validation Centre

Evaluate model predictions against independent observations using transparent, leakage-aware validation designs.

Worked example

Example: validate predicted seed set with grouped cross-validation by year or field and report MAE, RMSE, R-squared and calibration.

Tabs and views

Tab/view	Purpose
Data	Opens the data stage or result view.
Design	Opens the design stage or result view.
Results	Opens the results stage or result view.
Diagnostics	Opens the diagnostics stage or result view.
Export & project	Opens the export & project stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Validation-data source, Upload CSV containing one observed column and one or more prediction columns, Outcome type, Observed outcome.
3. Run “Run continuous validation”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Validation-data source	Single choice	Options: Upload paired observations and predictions, Use current Yield validation result, Use current Validation Centre data	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Upload CSV containing one observed column and	File upload	Accepted: csv	Uploads the file required for upload csv containing	Use the matching template supplied by

Field/control	Type	Options/default	Meaning	Example
one or more prediction columns			one observed column and one or more prediction columns. Check the displayed template or required column names before import.	AgroLattice.
Outcome type	Single choice	Options: Continuous outcome, Binary event	Select one option for “Outcome type”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Observed outcome	Drop-down	Options: generated from the current data or context	Select one option for “Observed outcome”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Prediction columns	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Prediction columns”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Evaluation design	Drop-down	Options: None, Grouped cross-validation, Temporal holdout, K-fold cross-validation Built-in help: None evaluates supplied predictions directly. Other options fit a simple linear calibration only on training partitions and evaluate it on held-out records.	Select one option for “Evaluation design”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Cross-validation folds	Slider	Range: 2 to 10 Default: 5	Sets the number of cross-validation partitions.	5 folds
Bootstrap iterations	Selection slider	Default: 1000	Enter or adjust the numeric setting “Bootstrap iterations”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Prediction threshold	Slider	Range: 0.0 to 1.0 Default: 0.5	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Bland–Altman model	Drop-down	Options: generated from the current data or context	Select one option for “Bland–Altman model”. Available choices may	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download validation CSV template	Download	Depends on the active data and prior selections.	Downloads the current validation csv template for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Run continuous validation	Action button	Depends on the active data and prior selections.	Runs the action named “Run continuous validation” using the current selections.	Review inputs, then run the action and inspect any warnings.
Run binary-event validation	Action button	Depends on the active data and prior selections.	Runs the action named “Run binary-event validation” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download validation research package	Download	Depends on the active data and prior selections.	Downloads the current validation research package for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Save validation snapshot to active project	Action button	Depends on the active data and prior selections.	Runs the action named “Save validation snapshot to active project” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

Use grouped validation whenever observations from the same field, year, trial or parent combination are related. Random row splitting can overstate performance.

10.2 Model ensemble & disagreement

Path: Models & Evidence > Model ensemble & disagreement

Combine selected model outputs while retaining inter-model spread, pairwise agreement and ranking instability.

Worked example

Example: average predictions from three models, quantify spread and inspect locations where the models disagree on ranking.

Tabs and views

Tab/view	Purpose
Data	Opens the data stage or result view.
Ensemble design	Opens the ensemble design stage or result view.
Predictions	Opens the predictions stage or result view.
Disagreement	Opens the disagreement stage or result view.
Export & project	Opens the export & project stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Ensemble-data source, Upload one row per record with one column per model, Observed outcome, when available, Model columns.
3. Run “Calculate ensemble and disagreement”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Ensemble-data source	Single choice	Options: Upload model-prediction CSV, Use Validation Centre predictions, Use current ensemble data	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Upload one row per record with one column per model	File upload	Accepted: csv	Uploads the file required for upload one row per record with one column per model. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Observed outcome, when available	Drop-down	Options: generated from the current data or context	Select one option for “Observed outcome, when available”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Model columns	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Model columns”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Ensemble method	Drop-down	Options: Mean, Median, Trimmed mean, Weighted mean, Inverse-error	Select one option for “Ensemble method”. Available choices may	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
		weighted	depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	
Record identifier	Drop-down	Options: generated from the current data or context	Select one option for "Record identifier". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Higher predictions indicate better outcomes	Checkbox	Default: True	Enables or disables "Higher predictions indicate better outcomes". Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
Calculate model consensus around a threshold	Checkbox	Depends on the active data and prior selections.	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Consensus threshold	Number	Default: 60.0	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Consensus condition	Single choice	Options: Above, Below	Select one option for "Consensus condition". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download ensemble CSV template	Download	Depends on the active data and prior selections.	Downloads the current ensemble csv template for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Calculate ensemble and disagreement	Action button	Depends on the active data and prior selections.	Runs the action named "Calculate ensemble and	Review inputs, then run the action and inspect any

Action	Type	Conditions/default	What it does	Good practice
			disagreement” using the current selections.	warnings.
Download ensemble research package	Download	Depends on the active data and prior selections.	Downloads the current ensemble research package for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Save ensemble snapshot to active project	Action button	Depends on the active data and prior selections.	Runs the action named “Save ensemble snapshot to active project” using the current selections.	Review inputs, then run the action and inspect any warnings.

Interpreting the result

A narrow ensemble spread suggests agreement, not necessarily correctness. Compare the ensemble with independent observations.

10.3 Yield validation

Path: Models & Evidence > Yield validation

Upload real yield records and test whether the app's yield, suitability, and water-stress indicators predict observed productivity.

Worked example

Example: upload observed maize yields by location and year and test whether the app's yield-screening index predicts them without data leakage.

Tabs and views

Tab/view	Purpose
Correlations	Opens the correlations stage or result view.
Model performance	Opens the model performance stage or result view.
Calibration plots	Opens the calibration plots stage or result view.
Feature importance	Opens the feature importance stage or result view.
Matched data	Opens the matched data stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Observed yield CSV, Geographic level of observed yields, App crop profile, Crop stage / seasonal window.
3. Run “Download interpretation as Markdown”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Observed yield CSV	File upload	Accepted: csv	Uploads the file required	Use the matching

Field/control	Type	Options/default	Meaning	Example
			for observed yield csv. Check the displayed template or required column names before import.	template supplied by AgroLattice.
Geographic level of observed yields	Single choice	Options: City and state, State only	Select one option for “Geographic level of observed yields”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
App crop profile	Drop-down	Options: generated from the current data or context	Select one option for “App crop profile”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Crop stage / seasonal window	Drop-down	Options: generated from the current data or context	Selects a crop-development phase such as establishment, flowering or maturity.	Flowering
State column	Drop-down	Options: generated from the current data or context	Select one option for “State column”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Year column	Drop-down	Options: generated from the current data or context	Select one option for “Year column”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
City column	Drop-down	Options: generated from the current data or context	Select one option for “City column”. Available choices may depend on the active country, dataset, project or	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	
Observed yield column	Drop-down	Options: generated from the current data or context	Select one option for "Observed yield column". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Crop column (optional)	Drop-down	Options: generated from the current data or context	Select one option for "Crop column (optional)". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Observed crop value to validate	Drop-down	Options: generated from the current data or context	Select one option for "Observed crop value to validate". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Months used for app crop predictors	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Selects explanatory variables supplied to a model.	GDD, water stress and sowing offset
Model random seed	Number	Range: 0 to 1000000 Default: 42	Fixed value used to reproduce a randomisation, clustering or model-fitting process.	2027
Cross-validated model prediction shown	Drop-down	Options: generated from the current data or context	Select one option for "Cross-validated model prediction shown". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			canopy sensitivity and reduce some background and atmospheric effects.	

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download Matched Validation Dataset CSV	Download	Depends on the active data and prior selections.	Downloads the current matched validation dataset csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download Cross-Validated Predictions CSV	Download	Depends on the active data and prior selections.	Downloads the current cross-validated predictions csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

10.4 Statistics toolbox

Path: Models & Evidence > Statistics toolbox

Filter the climate dataset and run descriptive, inferential, regression, time-series, and visual analyses.

Worked example

Example: filter temperature data to 1991-2025, calculate descriptive statistics, test correlations and fit a regression.

Tabs and views

Tab/view	Purpose
Descriptive	Opens the descriptive stage or result view.
Correlation	Opens the correlation stage or result view.
Hypothesis tests	Opens the hypothesis tests stage or result view.
Regression	Opens the regression stage or result view.
Time series	Opens the time series stage or result view.
Visualisations	Opens the visualisations stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Analysis level, Select variables, Select year range, Select months.

3. Run “Run Independent t-test”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Analysis level	Single choice	Options: Cities, States	Select one option for “Analysis level”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Select variables	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Select variables”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Select year range	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Enter or adjust the numeric setting “Select year range”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Select months	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Selects the months included in a crop window or climate summary.	May-September
Select {statistics_level.lower()}	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Select {statistics_level.lower()}”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Select statistics	Multi-select	Options: generated from the current data or context Default: count, mean, std, min, 25%, 50%, 75%, max	Select one or more entries for “Select statistics”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Group by	Multi-select	Options: generated from the current data or context Default: Variable, Month	Select one or more entries for “Group by”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Correlation method	Drop-down	Options: pearson, spearman	Select one option for “Correlation method”. Available choices may depend on the active	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	
High-correlation threshold	Slider	Range: 0.0 to 1.0 Default: 0.5	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Test type	Drop-down	Options: Independent t-test, Paired t-test, ANOVA	Select one option for “Test type”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Variable to test	Drop-down	Options: generated from the current data or context	Select one option for “Variable to test”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Group by	Drop-down	Options: generated from the current data or context, Year	Select one option for “Group by”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Group 1	Drop-down	Options: generated from the current data or context	Select one option for “Group 1”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Group 2	Drop-down	Options: generated from the current data or context	Select one option for “Group 2”. Available choices may depend on the active country, dataset, project or	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	
Select two years	Selection slider	Default: generated from the current data or context, generated from the current data or context	Enter or adjust the numeric setting “Select two years”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
ANOVA grouping variable	Drop-down	Options: generated from the current data or context, Year	Select one option for “ANOVA grouping variable”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Regression type	Drop-down	Options: Simple Linear, Multiple Linear	Select one option for “Regression type”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Dependent variable	Drop-down	Options: generated from the current data or context	Select one option for “Dependent variable”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Independent variable	Drop-down	Options: generated from the current data or context	Select one option for “Independent variable”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Independent variables	Multi-select	Options: generated from	Select one or more entries	Select only variables,

Field/control	Type	Options/default	Meaning	Example
		the current data or context Default: generated from the current data or context	for “Independent variables”. Selecting more entries broadens the analysis and may increase processing time.	locations or treatments required by the analysis.
Time-series variable	Drop-down	Options: generated from the current data or context	Select one option for “Time-series variable”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Time-series location	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Time-series method	Drop-down	Options: Trend Detection, Seasonality Decomposition, Autocorrelation	Select one option for “Time-series method”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Visualisation type	Drop-down	Options: Boxplot, Histogram, Line Plot	Select one option for “Visualisation type”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Variable to visualise	Drop-down	Options: generated from the current data or context	Select one option for “Variable to visualise”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Group or colour by	Drop-down	Options: None, Month, generated from the current data or context, Year	Select one option for “Group or colour by”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Compute Descriptive Statistics	Action button	Depends on the active data and prior selections.	Runs the action named "Compute Descriptive Statistics" using the current selections.	Review inputs, then run the action and inspect any warnings.
Download Descriptive Statistics CSV	Download	Depends on the active data and prior selections.	Downloads the current descriptive statistics csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Compute Correlation Matrix	Action button	Depends on the active data and prior selections.	Runs the action named "Compute Correlation Matrix" using the current selections.	Review inputs, then run the action and inspect any warnings.
Download Correlation Matrix CSV	Download	Depends on the active data and prior selections.	Downloads the current correlation matrix csv for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Run Independent t-test	Action button	Depends on the active data and prior selections.	Runs the action named "Run Independent t-test" using the current selections.	Review inputs, then run the action and inspect any warnings.
Run Paired t-test	Action button	Depends on the active data and prior selections.	Runs the action named "Run Paired t-test" using the current selections.	Review inputs, then run the action and inspect any warnings.
Run ANOVA	Action button	Depends on the active data and prior selections.	Runs the action named "Run ANOVA" using the current selections.	Review inputs, then run the action and inspect any warnings.
Run Simple Regression	Action button	Depends on the active data and prior selections.	Runs the action named "Run Simple Regression" using the current selections.	Review inputs, then run the action and inspect any warnings.
Run Multiple Regression	Action button	Depends on the active data and prior selections.	Runs the action named "Run Multiple Regression" using the current selections.	Review inputs, then run the action and inspect any warnings.
Run Time-Series Analysis	Action button	Depends on the active data and prior selections.	Runs the action named "Run Time-Series Analysis" using the current selections.	Review inputs, then run the action and inspect any warnings.
Generate Visualisation	Action button	Depends on the active data and prior selections.	Runs the action named "Generate Visualisation" using the current selections.	Review inputs, then run the action and inspect any warnings.
Download Plot as HTML	Download	Depends on the active data and prior selections.	Downloads the current plot as html for archiving,	Save it with the project, country and analysis date

Action	Type	Conditions/default	What it does	Good practice
			analysis or reporting.	in the filename.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

10.5 Scientific robustness

Path: Models & Evidence > Scientific robustness

Quantify uncertainty, audit data quality, correct multiple testing, and compare alternative analytical models.

Worked example

Example: bootstrap uncertainty, audit missingness, correct multiple tests and compare alternative model specifications.

Tabs and views

Tab/view	Purpose
Uncertainty	Opens the uncertainty stage or result view.
Data quality	Opens the data quality stage or result view.
Multiple testing	Opens the multiple testing stage or result view.
Model comparison	Opens the model comparison stage or result view.
City summary	Opens the city summary stage or result view.
Variable completeness	Opens the variable completeness stage or result view.
Duplicates	Opens the duplicates stage or result view.
Outliers	Opens the outliers stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Crop, Stage, Location, Period.
3. Run “Run data-quality audit”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Crop	Drop-down	Options: generated from the current data or context	Select the crop whose validated parameters or custom profile should be used.	Maize
Stage	Drop-down	Options: generated from the current data or context	Selects a crop-development phase such as establishment, flowering or maturity.	Flowering

Field/control	Type	Options/default	Meaning	Example
Location	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Period	Slider	Range: generated from the current data or context to generated from the current data or context Default: generated from the current data or context, generated from the current data or context	Defines the years or dates included in the calculation.	1991-2025
Months	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Selects the months included in a crop window or climate summary.	May-September
Aggregation	Single choice	Options: Limiting factor, Mean of components	Defines how multiple components are combined, for example by the limiting factor or their mean.	Limiting factor
Bootstrap / Monte Carlo runs	Slider	Range: 100 to 2000 Default: 500	Enter or adjust the numeric setting "Bootstrap / Monte Carlo runs". Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Threshold uncertainty (%)	Slider	Range: 0.0 to 25.0 Default: 5.0	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Random seed	Number	Default: 42	Fixed value used to reproduce a randomisation, clustering or model-fitting process.	2027
Variable	Drop-down	Options: generated from the current data or context	Select one option for "Variable". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
False-discovery threshold	Selection slider	Default: 0.05	Sets the value at which a condition, alert or classification changes.	Ks below 0.6
Comparison type	Single choice	Options: Climate similarity methods, Crop suitability aggregation	Select one option for "Comparison type". Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			and atmospheric effects.	
Reference city	Drop-down	Options: generated from the current data or context	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Year	Drop-down	Options: available years	Selects a calendar or season year included in the analysis.	2025
Top-N overlap	Slider	Range: 5 to 50 Default: 15	Enter or adjust the numeric setting “Top-N overlap”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download uncertainty simulations	Download	Depends on the active data and prior selections.	Downloads the current uncertainty simulations for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Run data-quality audit	Action button	Depends on the active data and prior selections.	Runs the action named “Run data-quality audit” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download complete quality audit ZIP	Download	Depends on the active data and prior selections.	Downloads the current complete quality audit zip for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download corrected trend tests	Download	Depends on the active data and prior selections.	Downloads the current corrected trend tests for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

11. Reports

This workspace packages results into reproducible reports, figures, tables, methods records and publication-ready exports.

Workspace selector

Open Reports from the sidebar. Use the Workspace view selector and, where present, the Analysis selector to move between related functions without leaving the workspace.

11.1 Study & publication builder

Path: Reports > Study & publication builder

Define a study protocol, select traceable outputs, create publication figures and tables, and export a manuscript package for author review.

Worked example

Example: select the trial protocol, synchrony metrics, validation table and figures, then export a manuscript package with a reproducibility manifest.

Tabs and views

Tab/view	Purpose
Study library	Opens the study library stage or result view.
Protocol	Opens the protocol stage or result view.
Tables	Opens the tables stage or result view.
Figures	Opens the figures stage or result view.
Build package	Opens the build package stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Load study, New study title, Title, Short title.
3. Run “Create new study”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Load study	Drop-down	Options: generated from the current data or context	Select one option for “Load study”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			some background and atmospheric effects.	
New study title	Text	Default: Integrated agroclimate modelling and validation in {selected_country}	Enter the requested text for “New study title”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Title	Text	Default: generated from the current data or context	Enter the requested text for “Title”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Short title	Text	Default: generated from the current data or context	Enter the requested text for “Short title”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Authors, one per line	Long text	Default: generated from the current data or context	Enter the requested text for “Authors, one per line”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Target journal	Text	Default: generated from the current data or context	Selects the response variable or event the analysis predicts or evaluates.	Seed-set percentage
Research question	Long text	Default: generated from the current data or context	Enter the requested text for “Research question”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Study design	Long text	Default: generated from the current data or context	Enter the requested text for “Study design”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Methods included	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Methods included”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Introduction draft	Long text	Default: generated from the current data or context	Enter the requested text for “Introduction draft”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Discussion draft	Long text	Default: generated from the current data or context	Enter the requested text for “Discussion draft”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Conclusion draft	Long text	Default: generated from the current data or context	Enter the requested text for “Conclusion draft”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Tables to include	Multi-select	Options: generated from the current data or	Select one or more entries for “Tables to include”.	Select only variables, locations or treatments

Field/control	Type	Options/default	Meaning	Example
		context Default: generated from the current data or context	Selecting more entries broadens the analysis and may increase processing time.	required by the analysis.
Figure data	Drop-down	Options: generated from the current data or context	Select one option for “Figure data”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Figure title	Text	Default: generated from the current data or context	Enter the requested text for “Figure title”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Figure caption	Long text	Depends on the active data and prior selections.	Enter the requested text for “Figure caption”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Load selected study	Action button	Depends on the active data and prior selections.	Runs the action named “Load selected study” using the current selections.	Review inputs, then run the action and inspect any warnings.
Create new study	Action button	Depends on the active data and prior selections.	Runs the action named “Create new study” using the current selections.	Review inputs, then run the action and inspect any warnings.
Generate publication figure	Action button	Depends on the active data and prior selections.	Runs the action named “Generate publication figure” using the current selections.	Review inputs, then run the action and inspect any warnings.
Clear generated figures	Action button	Depends on the active data and prior selections.	Performs a destructive or clearing action. Review confirmation controls and backups before using it. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Use only after verifying the active country, project, field or trial.
Build complete publication package	Action button	Depends on the active data and prior selections.	Runs the action named “Build complete publication package” using the current	Review inputs, then run the action and inspect any warnings.

Action	Type	Conditions/default	What it does	Good practice
			selections.	
Download publication package	Download	Depends on the active data and prior selections.	Downloads the current publication package for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

11.2 Workflow & reporting

Path: Reports > Workflow & reporting

Export reproducible reports, save presets, run batch jobs, inspect analysis history, generate transparent interpretations, and review complete methodology documentation.

Worked example

Example: save an analysis preset, export the complete analysis history and produce a methods-focused research report.

Tabs and views

Tab/view	Purpose
Reproducibility	Opens the reproducibility stage or result view.
Research reports	Opens the research reports stage or result view.
Saved presets	Opens the saved presets stage or result view.
Batch analysis	Opens the batch analysis stage or result view.
Analysis history	Opens the analysis history stage or result view.
Automated interpretation	Opens the automated interpretation stage or result view.
Methods documentation	Opens the methods documentation stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Report title, Research question / notes, Results included, Preset name.
3. Run “Download HTML research report”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Report title	Text	Default: AgroLattice Research Report	Enter the requested text for “Report title”. Use consistent identifiers and document methods or	Use a clear, stable name that will remain meaningful in exports.

Field/control	Type	Options/default	Meaning	Example
			sources where relevant.	
Research question / notes	Long text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Results included	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Select one or more entries for “Results included”. Selecting more entries broadens the analysis and may increase processing time.	Select only variables, locations or treatments required by the analysis.
Preset name	Text	Depends on the active data and prior selections.	Enter the requested text for “Preset name”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
Description	Long text	Depends on the active data and prior selections.	Stores contextual information, methods, data provenance or interpretation notes.	Include instrument, sampling method and citation.
Preset to apply	Drop-down	Options: generated from the current data or context	Select one option for “Preset to apply”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Batch request CSV	File upload	Accepted: csv	Uploads the file required for batch request csv. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Result to interpret	Drop-down	Options: generated from the current data or context	Select one option for “Result to interpret”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Copy-ready interpretation	Long text	Default: generated from the current data or context	Enter the requested text for “Copy-ready interpretation”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download reproducibility metadata JSON	Download	Depends on the active data and prior selections.	Downloads the current reproducibility metadata json for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download HTML research report	Download	Depends on the active data and prior selections.	Downloads the current html research report for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download complete report ZIP	Download	Depends on the active data and prior selections.	Downloads the current complete report zip for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Apply selected preset	Action button	Depends on the active data and prior selections.	Runs the action named “Apply selected preset” using the current selections.	Review inputs, then run the action and inspect any warnings.
Delete selected preset	Action button	Depends on the active data and prior selections.	Performs a destructive or clearing action. Review confirmation controls and backups before using it. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Use only after verifying the active country, project, field or trial.
Download presets JSON	Download	Depends on the active data and prior selections.	Downloads the current presets json for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download batch-analysis template	Download	Depends on the active data and prior selections.	Downloads the current batch-analysis template for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Run batch analysis	Action button	Depends on the active data and prior selections.	Runs the action named “Run batch analysis” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download batch results	Download	Depends on the active data and prior selections.	Downloads the current batch results for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download analysis history	Download	Depends on the active data and prior selections.	Downloads the current analysis history for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Clear stored history	Action button	Depends on the active data and prior selections.	Performs a destructive or clearing action. Review confirmation controls and backups before using it. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background	Use only after verifying the active country, project, field or trial.

Action	Type	Conditions/default	What it does	Good practice
			and atmospheric effects.	

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

12. Data & Settings

This workspace manages country datasets, NASA acquisition, projects, crop libraries, backups, AI interpretation and system diagnostics.

Workspace selector

Open Data & Settings from the sidebar. Use the Workspace view selector and, where present, the Analysis selector to move between related functions without leaving the workspace.

12.1 Country workspace

Path: Data & Settings > Country workspace

Shows whether the selected country has an installed climate dataset and provides a searchable country map and storage information.

Worked example

Example: select Cyprus in the sidebar, confirm that the Cyprus dataset is installed, then use map search to locate Athalassa or another research site.

Typical workflow

1. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

This view is primarily informational or inherits its controls from the global sidebar and active workspace context.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

12.2 Dataset updater

Path: Data & Settings > Dataset updater

Select any country represented in worldcities.csv, download NASA POWER point data for its locations, aggregate to monthly long format, audit it, and install it into an independent country workspace.

Worked example

Example: select Cyprus, choose Nicosia and Limassol, download NASA data for 1985-2025, audit coverage, finalise the candidate and install it into the Cyprus country folder.

Tabs and views

Tab/view	Purpose
1 - Configure	Opens the 1 - configure stage or result view.
2 - Download batches	Opens the 2 - download batches stage or result view.
3 - Finalise & install	Opens the 3 - finalise & install stage or result view.
4 - Audit & methods	Opens the 4 - audit & methods stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Country to download or update, Start date, End date, Include partial edge months.
3. Run “Create update job”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Country to download or update	Drop-down	Options: generated from the current data or context Built-in help: NASA POWER is global. This selection controls the location catalogue, country dataset, updater checkpoint, cache and backups.	Selects the country context or country-specific dataset.	Mexico
Start date	Date	Default: generated from the current data or context Built-in help: The default is the first day after the latest month in the active dataset.	Defines the temporal boundary of the analysis or data request.	1 May 2027
End date	Date	Default: generated from the current data or context Built-in help: The default excludes the current month and applies a short source-availability lag.	Defines the temporal boundary of the analysis or data request.	1 May 2027
Include partial edge months	Checkbox	Default: False Built-in help: Not recommended for the main historical dataset. A partial first or final month can bias comparisons.	Selects the months included in a crop window or climate summary.	May-September
Location scope	Drop-down	Options: generated from the current data or context Built-in help: For a new country, start with selected cities or regions. Downloading every city can create a very large dataset and many NASA requests.	Select one option for “Location scope”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Administrative regions	Multi-select	Options: generated from the current data or context Default: generated from the current data or context	Selects a first-level administrative area such as a state, province or district.	Guanajuato
Cities	Multi-select	Options: generated from the current data or	Select one or more entries for “Cities”. Selecting	Select only variables, locations or treatments

Field/control	Type	Options/default	Meaning	Example
		context Default: generated from the current data or context	more entries broadens the analysis and may increase processing time.	required by the analysis.
Deduplicate locations by approximate NASA meteorological grid cell	Checkbox	Default: True Built-in help: Recommended. Cities resolving to the same approximate 0.5° x 0.625° meteorological cell share one request, then receive separate city records.	Downloads one NASA point for cities that resolve to the same approximate grid cell, then assigns the shared profile to each city record.	Enabled
Request legacy optional NASA parameters	Checkbox	Default: False Built-in help: Adds EVLAND, CLOUD_AMT_DAY and EVPTRNS. The updater falls back to the core set if POWER rejects them.	Requests optional variables used by older datasets. Leave off unless compatibility is required.	Disabled
Minimum monthly daily coverage	Slider	Range: 70 to 100 Default: 90	Minimum percentage of expected daily values required before a monthly aggregate is accepted.	90%
Delay between grid requests (seconds)	Number	Range: 0.0 to 30.0 Default: 1.5 Built-in help: A short delay reduces the risk of overloading or being rate-limited by the public API.	Pauses between public NASA requests to reduce rate-limiting risk.	1.5 seconds
How should downloaded rows be merged?	Single choice	Options: Fill missing keys only — preserve all existing values, Replace matching keys in the selected period — refresh NASA values	Select one option for “How should downloaded rows be merged?”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Allow reset of existing job	Checkbox	Default: False	Enables or disables “Allow reset of existing job”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.
Grid cells in this batch	Number	Range: 1 to generated from the current data or context Default: generated from the current data or context	Enter or adjust the numeric setting “Grid cells in this batch”. Use documented units and a scientifically justified value.	Start with the default, then test sensitivity where appropriate.
Ignore local NASA cache	Checkbox	Default: False	Forces a new request instead of reusing a compatible locally cached response.	Disabled
Process all remaining grids now	Checkbox	Default: False Built-in help: This may keep the Streamlit page occupied for a long time.	Enables or disables “Process all remaining grids now”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.

Field/control	Type	Options/default	Meaning	Example
Create mixed-format export	Checkbox	Default: False	Creates an optional legacy-style export. The current app uses the long-format country dataset, so this can normally remain off.	Disabled
Create legacy very-wide export	Checkbox	Default: False Built-in help: The wide file can be extremely large and is not required by the current app.	Creates a large very-wide legacy file. It is not required by the current app.	Disabled
I have reviewed the candidate and want to replace the active dataset	Checkbox	Default: False	Enables or disables “I have reviewed the candidate and want to replace the active dataset”. Leave it at the default unless the alternative is intentional. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Toggle only when the displayed workflow requires it.
Delete the active checkpoint, staging rows, candidate and audit files	Checkbox	Default: False	Enables or disables “Delete the active checkpoint, staging rows, candidate and audit files”. Leave it at the default unless the alternative is intentional.	Toggle only when the displayed workflow requires it.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Create update job	Action button	Depends on the active data and prior selections.	Runs the action named “Create update job” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download next batch	Action button	Depends on the active data and prior selections.	Runs the action named “Download next batch” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download staged update rows	Download	Depends on the active data and prior selections.	Downloads the current staged update rows for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Finalise candidate dataset	Action button	Depends on the active data and prior selections.	Runs the action named “Finalise candidate dataset” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download complete updated dataset package	Download	Depends on the active data and prior selections.	Downloads the current complete updated dataset package for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Install candidate and reload app	Action button	Depends on the active data and prior selections.	Runs the action named “Install candidate and reload app” using the current selections.	Review inputs, then run the action and inspect any warnings.

Action	Type	Conditions/default	What it does	Good practice
Download request log	Download	Depends on the active data and prior selections.	Downloads the current request log for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download candidate validation audit	Download	Depends on the active data and prior selections.	Downloads the current candidate validation audit for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download {heading.lower()}	Download	Depends on the active data and prior selections.	Downloads the current {heading.lower()} for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download complete coverage audit	Download	Depends on the active data and prior selections.	Downloads the current complete coverage audit for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Reset updater workspace	Action button	Depends on the active data and prior selections.	Performs a destructive or clearing action. Review confirmation controls and backups before using it. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Use only after verifying the active country, project, field or trial.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

12.3 Validated crop parameters

Path: Data & Settings > Validated crop parameters

Inspect, audit and download the literature-derived defaults used by crop suitability, irrigation and water-stress analyses.

Worked example

Example: inspect the maize temperature envelope, crop coefficients and root-depth assumptions before running suitability or water balance analyses.

Tabs and views

Tab/view	Purpose
Climate envelope	Opens the climate envelope stage or result view.
Stage Kc / Ky	Opens the stage kc / ky stage or result view.
Sources	Opens the sources stage or result view.
Replacement rules	Opens the replacement rules stage or result view.
Evidence gaps	Opens the evidence gaps stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Crop, Crop profile.
3. Run “Download interpretation as Markdown”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Crop	Drop-down	Options: generated from the current data or context	Select the crop whose validated parameters or custom profile should be used.	Maize
Crop profile	Drop-down	Options: generated from the current data or context	Select one option for “Crop profile”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Download interpretation as Markdown	Download	Depends on the active data and prior selections.	Downloads the current interpretation as markdown for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

12.4 Data exchange, backup & audit

Path: Data & Settings > Data exchange, backup & audit

Create offline field packs, re-import field observations and task updates, preserve an audit trail and export an API-ready data bundle.

Worked example

Example: create an offline field pack before a site visit, enter observations without internet, then re-import the completed CSVs and preserve the audit trail.

Tabs and views

Tab/view	Purpose
Offline field pack	Opens the offline field pack stage or result view.
Offline import	Opens the offline import stage or result view.
Backup & integrations	Opens the backup & integrations stage or result view.
Audit log	Opens the audit log stage or result view.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: Field, Completed observation_import_template.csv, Completed task_status_updates.csv, Completed operation_import_template.csv.
3. Run “Import offline observations”.
4. Review warnings, assumptions, provenance and the main table, chart or map.
5. Download the result or complete package for reproducibility.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
Field	Drop-down	Options: generated from available data	Selects the geographic or experimental unit used by the analysis.	A mapped research field or catalogue city
Completed observation_import_template.csv	File upload	Accepted: csv	Uploads the file required for completed observation_import_template.csv. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Completed task_status_updates.csv	File upload	Accepted: csv	Uploads the file required for completed task_status_updates.csv. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.
Completed operation_import_template.csv	File upload	Accepted: csv	Uploads the file required for completed operation_import_template.csv. Check the displayed template or required column names before import.	Use the matching template supplied by AgroLattice.

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Import offline observations	Action button	Depends on the active data and prior selections.	Runs the action named “Import offline observations” using the current selections.	Review inputs, then run the action and inspect any warnings.
Import task updates	Action button	Depends on the active	Runs the action named	Review inputs, then run

Action	Type	Conditions/default	What it does	Good practice
		data and prior selections.	“Import task updates” using the current selections.	the action and inspect any warnings.
Import field operations	Action button	Depends on the active data and prior selections.	Runs the action named “Import field operations” using the current selections.	Review inputs, then run the action and inspect any warnings.
Download complete operations backup	Download	Depends on the active data and prior selections.	Downloads the current complete operations backup for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.
Download integration schema JSON	Download	Depends on the active data and prior selections.	Downloads the current integration schema json for archiving, analysis or reporting.	Save it with the project, country and analysis date in the filename.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

12.5 ChatGPT settings

Path: Data & Settings > ChatGPT settings

Configure the OpenAI API connection, reasoning budget, and style of the human-readable scientific reports generated throughout the app.

Worked example

Example: enter a server-managed API key, choose a concise scientific style and limit the number of result rows sent for interpretation.

Typical workflow

1. Select the active country, project, field or trial context required by the tool.
2. Configure the main inputs: OpenAI API key, OpenAI model, Maximum output tokens, Reasoning effort.
3. Run “Test API connection”.
4. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

Some controls are conditional and appear only after a related option, tab, dataset or trial has been selected.

Field/control	Type	Options/default	Meaning	Example
OpenAI API key	Text	Default: generated from the current data or context Built-in help: A key entered here overrides OPENAI_API_KEY for this session. It is not saved to disk by this app.	Enter the requested text for “OpenAI API key”. Use consistent identifiers and document methods or sources where relevant.	Use a clear, stable name that will remain meaningful in exports.
OpenAI model	Text	Built-in help: Use gpt-5.6 for GPT-5.6 Sol, or another Responses API-compatible model available to your	Enter the requested text for “OpenAI model”. Use consistent identifiers and document methods or	Use a clear, stable name that will remain meaningful in exports.

Field/control	Type	Options/default	Meaning	Example
		OpenAI project.	sources where relevant.	
Maximum output tokens	Number	Range: 1000 to 20000 Built-in help: Reasoning tokens and the final report share this allowance. 6,000 is a reliable starting point for research reports.	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value
Reasoning effort	Drop-down	Options: none, low, medium, high, xhigh, max Built-in help: Low is recommended for result interpretation. Higher settings can use more output tokens before the final report is written.	Select one option for “Reasoning effort”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Text verbosity	Drop-down	Options: low, medium, high Built-in help: Controls the default level of detail in GPT-5.6 text output.	Select one option for “Text verbosity”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Detail level	Drop-down	Options: Concise, Standard, Detailed	Select one option for “Detail level”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Primary audience	Drop-down	Options: Researcher, Agronomist, Policy maker, General reader	Select one option for “Primary audience”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Choose the option matching the research question.
Report language	Drop-down	Options: English, Spanish	Select one option for “Report language”. Available choices may depend on the active country, dataset, project or previous selection. Terms: EVI: Enhanced	Choose the option matching the research question.

Field/control	Type	Options/default	Meaning	Example
			Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	
Maximum result rows sampled for each interpretation	Slider	Range: 10 to 100 Built-in help: The app sends descriptive summaries plus a bounded head/tail sample, not the full climate dataset.	Sets a lower or upper screening, processing or validation limit.	Use a locally justified value

Actions and downloads

Action	Type	Conditions/default	What it does	Good practice
Test API connection	Action button	Depends on the active data and prior selections.	Runs the action named “Test API connection” using the current selections.	Review inputs, then run the action and inspect any warnings.
Clear session API key	Action button	Depends on the active data and prior selections.	Performs a destructive or clearing action. Review confirmation controls and backups before using it. Terms: EVI: Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.	Use only after verifying the active country, project, field or trial.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

12.6 System diagnostics

Path: Data & Settings > System diagnostics

Checks whether datasets, optional dependencies and current-session analytical products are available and records the last page error.

Worked example

Example: use diagnostics after a failed satellite request to confirm whether the satellite libraries are installed and inspect the latest stored error.

Typical workflow

1. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

This view is primarily informational or inherits its controls from the global sidebar and active workspace context.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

13. Help

This workspace explains the application, feature boundaries, terminology and release history.

Workspace selector

Open Help from the sidebar. Use the Workspace view selector and, where present, the Analysis selector to move between related functions without leaving the workspace.

13.1 Quick guide & glossary

Path: Help > Quick guide & glossary

Provides an in-app orientation, workflow hints and a compact glossary.

Worked example

Example: look up “RAW”, “NDMI” or “anthesis” before configuring an analysis.

Typical workflow

1. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

This view is primarily informational or inherits its controls from the global sidebar and active workspace context.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

13.2 Commercial feature benchmark

Path: Help > Commercial feature benchmark

Explains which commercial-style capabilities are implemented locally and which require cloud services, hardware integrations or crop-specific validation.

Worked example

Example: confirm that automatic machinery control and secure cloud authentication are not provided by the local desktop release.

Typical workflow

1. Review warnings, assumptions, provenance and the main table, chart or map.

Fields and selections

This view is primarily informational or inherits its controls from the global sidebar and active workspace context.

Interpreting the result

Interpret the output together with the active country, location, time period, crop profile, data completeness, model assumptions and any warnings displayed by the page.

14. Key formulas and scientific conventions

14.1 Growing degree days

$$\text{Daily GDD} = \max[0, ((T_{\min}^* + T_{\max}^*) / 2) - T_{\text{base}}]$$

T_{min} and T_{max}* are constrained between the base temperature and upper cap.*

With a 10 °C base and 30 °C cap, a day with T_{min} = 8 °C and T_{max} = 26 °C contributes ((10 + 26) / 2) - 10 = 8 GDD.

14.2 Crop evapotranspiration

$$ET_c = K_c \times ET_0$$

Crop water use is estimated by multiplying reference evapotranspiration by the crop coefficient.

14.3 Root-zone available water

$$TAW = 1000 \times (\theta_{FC} - \theta_{WP}) \times \text{root depth}$$

TAW is in millimetres when root depth is in metres and water contents are volumetric fractions.

$$RAW = p \times TAW$$

RAW is the readily available part of total available water.

14.4 Maize seed set

$$\text{Seed set (\%)} = 100 \times \text{filled kernels} / (\text{filled kernels} + \text{unfilled kernels})$$

Use consistent counting rules and define the sampled ears and kernel sites.

14.5 Synchrony gap

$$\text{Synchrony gap} = \text{male 50\% pollen-shed date} - \text{female 50\% silking date}$$

A positive value means male 50% pollen shed occurred later than female 50% silking.

14.6 Satellite indices

$$NDVI = (NIR - Red) / (NIR + Red)$$

Higher NDVI often indicates more green vegetation, but interpretation depends on crop stage and canopy conditions.

$$NDMI = (NIR - SWIR) / (NIR + SWIR)$$

NDMI is sensitive to canopy moisture-related variation.

14.7 Prediction metrics

$$MAE = \text{mean}(|\text{observed} - \text{predicted}|)$$

Average error magnitude in the original outcome units.

$$RMSE = \sqrt{\text{mean}((\text{observed} - \text{predicted})^2)}$$

Gives more weight to large prediction errors.

15. Agricultural and analytical glossary

Term	Brief explanation
AOI	Area of interest: the exact point buffer or polygon analysed on a map or by satellite.
Anthesis	The flowering stage when anthers release pollen. In maize, it is commonly represented by active pollen shed from the male parent.
Silking	Emergence of silks from the female maize ear. Silks must be receptive while viable pollen is available.
Detasselling	Removal of tassels from female maize rows before pollen release to prevent self-pollination in hybrid seed production.
Seed set	The percentage of potential kernel sites that become filled kernels after successful fertilisation and development.
Pure seed	The proportion of a sample that meets the programme's physical or varietal definition of pure seed. The exact laboratory definition must be documented.
Genetic purity	The proportion of seed that genetically matches the intended hybrid or parentage, measured using an accepted purity method.
Kernel rows per ear	The number of longitudinal rows of kernels around a maize ear; it is a yield-component and ear-trait measurement.
GDD	Growing degree days: accumulated thermal time above a base temperature, usually capped at an upper temperature.
Phenology	The timing of crop development stages such as emergence, flowering and maturity.
ET ₀	Reference evapotranspiration: atmospheric evaporative demand from a standard reference surface.
ET _c	Crop evapotranspiration: estimated crop water use, commonly calculated as crop coefficient K _c multiplied by ET ₀ .
K _c	Crop coefficient: a stage-dependent multiplier that converts ET ₀ into crop evapotranspiration.
TAW	Total available water: root-zone water held between field capacity and permanent wilting point.
RAW	Readily available water: the fraction of TAW that can be depleted before crop water stress is expected.
K _s	Water-stress coefficient: a value near 1 indicates little water limitation; lower values reduce estimated crop evapotranspiration.
Field capacity	The water content remaining after excess gravitational water has drained from the soil.
Permanent wilting point	The soil water content below which plants generally cannot recover turgor.
Root-zone depletion	The amount of plant-available water removed from the effective rooting depth relative to field capacity.
Deep percolation	Water that drains below the simulated root zone and is no longer available to the crop in the model.
Capillary rise	Upward movement of groundwater into the root zone; it should be used only when supported by local evidence.
Curve number	An NRCS runoff parameter representing rainfall-runoff potential; higher values generally imply more runoff.

Term	Brief explanation
NDVI	Normalised Difference Vegetation Index, based on red and near-infrared reflectance; commonly used as a canopy greenness/vigour indicator.
EVI	Enhanced Vegetation Index, designed to improve canopy sensitivity and reduce some background and atmospheric effects.
NDMI	Normalised Difference Moisture Index, sensitive to canopy water content and moisture-related change.
NDRE	Normalised Difference Red Edge Index, often sensitive to chlorophyll and canopy status in developed crops.
SCL	Sentinel-2 Scene Classification Layer, used to mask clouds, shadows, snow and other unwanted pixel classes.
PCA	Principal component analysis: a dimension-reduction method that summarises correlated climate variables into orthogonal axes.
K-means	A clustering algorithm that partitions observations around cluster centres in feature space.
Silhouette score	A cluster-quality measure comparing within-cluster cohesion with separation from other clusters.
MAE	Mean absolute error: the average absolute difference between prediction and observation.
RMSE	Root mean squared error: an error metric that gives greater weight to large errors.
R-squared	The proportion of observed variance explained by predictions; it can be negative on validation data.
Calibration	Agreement between predicted and observed levels across the prediction range.
Grouped cross-validation	Validation that keeps related observations, such as the same field or year, together to reduce leakage.
NASA POWER	A global gridded meteorological and solar-data service used by the app for point-based climate and daily weather acquisition.
Sentinel-2	A European optical Earth-observation satellite mission providing multispectral imagery, including 10 m and 20 m bands.
AquaCrop	FAO's crop-water productivity model, represented here through AquaCrop-OSPy when installed.
DSSAT	Decision Support System for Agrotechnology Transfer, an external process-based crop-modelling system.
APSIM	Agricultural Production Systems sIMulator, an external process-based modelling framework.

15.1 Additional terms

Term	Brief explanation
Administrative region	A state, province, district or comparable first-level geographic unit used to filter the country catalogue.
Climate analogue	A historical year or location with a climate profile resembling the target condition.
Climate envelope	A set of temperature, rainfall or water-balance ranges used to

Term	Brief explanation
	screen crop suitability.
Crop stage	A defined phase of development, such as establishment, vegetative growth, flowering or maturity.
Effective precipitation	The portion of rainfall assumed to be stored and usable by the crop after runoff, interception and deep drainage.
Flowering overlap	The amount of time or weighted area during which male pollen shed and female silk receptivity occur together.
Hybrid seed production	Production in which a designated female parent is fertilised by a designated male parent to create hybrid seed.
Management zone	A field subdivision with similar measured characteristics that may receive a common management rate.
Planting window	A range of possible planting dates evaluated for climate or crop suitability.
Potential yield	A user-supplied or modelled yield ceiling under the defined assumptions, used to scale a relative index.
Randomised block design	A trial design in which all treatments are randomised within spatial blocks to control field heterogeneity.
Receptive silks	Silks that are biologically capable of receiving pollen and supporting fertilisation.
Remote sensing	Observation of the field using satellite or airborne measurements rather than direct contact.
Sowing treatment	An experimentally assigned planting date or offset used to compare flowering synchrony or crop performance.
Thermal time	Accumulated temperature units used to represent biological development more consistently than calendar days.
Variable-rate prescription	A spatial file assigning different application rates to different parts of a field.

16. Troubleshooting and maintenance

Problem	Recommended action
App does not launch	Run DIAGNOSE_APP.bat. Confirm that agrolattice.py exists and Streamlit is installed in the active Python environment.
Country has no climate data	Open Dataset updater for that country, create or resume a job, finalise the candidate and install it.
Wrong location coordinates	Correct Datasets\worldcities.csv, reset any updater job created with the old coordinates, ignore the NASA cache and redownload the affected location and period.
NASA download fails	Retry later, reduce batch size, keep a delay between requests, inspect the request log and use checkpoints rather than restarting the whole job.
Satellite STAC fails	Use Automatic failover, retry later and check satellite dependencies. Public catalogues can return temporary gateway errors.
Satellite results belong to another field	Clear the current AOI or resend the exact field/plot geometry. AgroLattice uses geometry hashes to prevent mismatched attachments.
Clustering fails or collapses	Reduce the requested clusters. Small countries may contain duplicate NASA grid profiles and fewer distinct climate profiles than locations.
No Twin state	Link a field, load weather and optional root-zone/satellite/sensor data, then open Twin configuration to inspect readiness.
Model performance looks unrealistically high	Use grouped validation by field, trial, year or parent combination and remove post-outcome leakage variables.
HTML code appears as text	Confirm that ui_release4.py and the main app are from the same release and clear __pycache__ before restarting.

16.1 Backup and cleanup

- Keep a clean release ZIP outside the working folder.
- Keep at least one recent country-dataset backup until analyses are verified.
- Completed updater active_job folders can be reset after successful installation and validation.
- __pycache__ can be deleted while the app is closed; Python recreates it.
- Do not delete pollination_lab, field_operations, agrolattice_twin, project_store or study_store unless the associated records are intentionally being removed.
- Before changing paths or updating versions, close Streamlit and any program that has the dataset open.

16.2 Reproducibility checklist

- Record app version and country dataset checksum.
- Record location, coordinates, field geometry and geometry hash.
- Record crop profile, GDD base/cap, soil assumptions and irrigation settings.
- Save raw observations and imported files unchanged.
- Use fixed random seeds for randomisation and modelling.
- Use grouped validation and report uncertainty.
- Download complete packages, not only screenshots.
- Document manual edits, exclusions and protocol deviations.